



Crash Injury Research and Engineering Network (CIREN)

Data Dictionary (version 1.2.1, April 3, 2017)

2016 Data Release

[Crash dates June 1, 2005 through December 31, 2016]

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Section 1 – Introduction

The Crash Injury Research and Engineering Network (CIREN) is a multidisciplinary program dedicated to the study of injury causation associated with motor vehicle crashes. CIREN was designed on the recommendations put forth by the National Academy of Sciences (NAS) in their book entitled “Injury in America – A Continuing Public Health Problem”¹. The recommendations called for a coordinated approach to multidisciplinary biomechanics research that included:

- Injury investigation, injury mechanism study, biomedical research on response and tolerance, study of pathophysiologic response to impact, and research on disabling injuries
- Support for the training of scientists and engineers in injury biomechanics to overcome a serious shortage of such workers
- Support of incentives for established investigators on university faculties to develop curricula in the mechanics and physiology of injury
- Development and nurturing of a core of leaders by establishing university centers for scientific study of injury and disability; adequate support for long-term and applied research should be emphasized and an effort made to curb the rapid loss of trained researchers primarily to service as expert witnesses in lawsuits
- Maintenance of a balanced extramural research program managed by engineers, physiologists, and physicians
- Fostering of cooperation and exchange of information with existing federal organizations
- Promotion of the development of information on important problems in trauma, independent of the regulatory function of the government

In 1997, CIREN was developed as an extension of a program called “Highway Traffic Injury Studies” which was NHTSA’s original response to the NAS recommendations. The Hospital Studies looked at the in-depth side of injury related to motor vehicle crashes at four large level 1 trauma centers. However, there was a need to grow the academic and engineering portion of the crash investigations and injury causation research. As NHTSA continued to develop CIREN, additional expertise in the areas of engineering, epidemiology, and traumatology were added. The primary focus of CIREN case recruitment has consistently remained on crashes involving late model vehicles resulting in serious and/or disabling injury. In 2000, the electronic database was completed for the collection and storage of CIREN data.

The CIREN electronic data system shares much of its core structure with the National Automotive Sampling System – Crashworthiness Data System (NASS-CDS). The entire vehicle and scene data structure in CIREN is exactly the same as in NASS-CDS. The CIREN crash investigators receive the same training and utilize the same techniques as those used in NASS-CDS for vehicle and scene documentation. CIREN and NASS-CDS share the same data schema for the vehicle and scene data collected.

The detailed injury-focused data captured by CIREN required a unique data schema for the occupant-based variables. The CIREN schema captures extensive data related to occupant pre-crash health status, diagnostic procedures and results, invasive procedures, anthropometric data, injury data, and outcomes. It is the review by the multidisciplinary CIREN teams of the scene, vehicle, and occupant data that then allows CIREN to generate its final data step known as BioTab. The BioTab (short for Biomechanics Tab) was established in 2005 and is a method to objectively and completely analyze and document the causes of injuries using data obtained from detailed medical records and imaging, in-depth crash investigations, and findings from the medical and

biomechanical literature². BioTab contains Injury Causation Scenarios (ICS) that describe the Involved Physical Component(s) (IPC) and supporting evidence such as injury patterns, contact points, and safety system interactions for each serious and disabling injury sustained. In addition, the BioTab captures relevant biomechanics data such as load paths, sources of energy, and mechanisms of injury. Load paths are coded based on a 20-body-region design that allows for the breakout of the joints and segments of the human anatomy. This allows for a greater level of specificity compared to the more traditional eight body regions utilized by the Abbreviated Injury Scale (AIS)^{3,4}. Injury mechanisms are also coded in the context of these 20 body regions. The BioTab allows for a primary and secondary regional mechanism, which describe the mechanical loading applied to the entire region, to be coded for each injury. Another unique feature of BioTab is the assignment and coding of contributing factors. Contributing factors are factors that increase the severity or likelihood of sustaining the injury being coded. An example of a contributing factor would be a comorbidity of osteoporosis for an occupant who sustains a significant fracture in an event that would otherwise not be expected to produce a fracture of that severity. Contributing factors are selected based on expert review and consensus. Lastly, all ICS(s) and IPC(s) are given a confidence level based on the evidence available from the field investigation, hospital course, physician review, engineering review, and literature support. The data set generated for the CIREN public release will come entirely from the BioTab coding era (mid-2005 and later) to provide users with the most comprehensive injury causation data available.

The CIREN case review is the keystone of the entire program since it is built on the fundamental recommendations originally set forth by the National Academy of Sciences. The multidisciplinary review incorporates all aspects of the CIREN case and promotes a rigorous discussion involving crash investigators, engineers, physicians, researchers and other experts including NHTSA. Expert consensus on injury causation coding is achieved by presentation and review of the crash evidence (vehicle and scene), occupant data (injury, medical history, radiology, invasive procedures, and outcomes), medical and biomechanical literature, and a review of relevant crash test data.

This CIREN Data Dictionary has been assembled to inform researchers on the variables, attributes, and common key fields in the CIREN data files. This manual will also describe appropriate uses of CIREN data.

Section 2 – CIREN Design

Although CIREN shares much of its data design and data collection with NASS-CDS, it is not a probability sample of all police reported crashes in the U.S. like NASS-CDS. CIREN is a convenience sample of crash occupants that are admitted to a CIREN-participating trauma center, meet the CIREN inclusion criteria, and agree to the Institutional Review Board-approved protocol via informed consent.

NASS-CDS cases incorporate all the occupants and vehicles within a particular crash, whereas a CIREN case is an individual occupant within a crash. If multiple occupants are involved in a crash, it is possible for some occupants to be captured in CIREN while others may not be. Reasons why other occupants may not be enrolled include lack of serious or disabling injury or because the occupant refused informed consent.

CIREN inclusion criteria are based on two different occupant populations: adult and pediatric. Pediatric cases have typically been classified as those occupants 12 years of age or younger, while adult cases are occupants 13 years old and above. Although small changes to the CIREN inclusion criteria have been made over the years, the core requirements have remained intact. The primary elements for CIREN inclusion are:

- Occupants sustaining a serious or disabling injury based on the AIS
 - Pediatric cases have a lower threshold of moderate severity
- Vehicles should be no more than 6 years old from the current manufacturing year
 - Pediatric cases allow a vehicle age of 8 years
- Sisters and clone vehicles are allowed beyond the model year limit if significant vehicle changes have not been made
- Crash types allowed include frontal, side (near and far), and rollover
 - Pediatric cases can be in rear-impact type crashes
 - Single significant event crashes are preferred

CIREN data are for case study analysis. The data are not intended nor designed to be used for national estimates or risk analyses. The CIREN data are designed to focus on the details of injury causation and injury mechanisms resulting from motor vehicle crashes.

Section 3 – NASS-CDS variables (vehicle and scene)

The NASS-CDS data schema is duplicated in CIREN for the collection of vehicle and scene data. Therefore, the “NASS-CDS Analytical User’s Manual” should be referenced for appropriate interpretation of the NASS-CDS vehicle and scene variables that are captured for CIREN cases.

The NASS-CDS data is completed on an annual basis and made available for public use in data sets based on the year of the crash. The NASS-CDS Analytical User’s Manual is updated annually to correspond with the release of each year’s NASS-CDS data. The CIREN public data set is different in that multiple years of data are combined into the one data set release. The 2016 CIREN data release will cover published cases with crash dates from June 1, 2005 until December 31, 2016. The “Publish” status for a CIREN case indicates the case has completed the review and quality control stages of development and is ready for public dissemination. Future releases will include additional cases from this era as they are published and also add published cases from more recent years. Using the CIREN public data set appropriately will require this manual for CIREN-specific data and the appropriate NASS-CDS Analytical User’s Manual(s) for the vehicle and scene data from respective crash dates in the applicable CIREN cases (i.e. a 2009 CIREN case year requires use of the 2009 Analytical User’s Manual to find information about vehicle and scene variables). Cases enrolled in 2016 follow 2015 guidelines.

Electronic copies of the NASS-CDS Analytical User’s Manual can be found under the “Manuals and Documentation” link on the NASS-CDS webpage located in the Data section of the NHTSA web site.

(current URL - <http://www-nrd.nhtsa.dot.gov/cats/listpublications.aspx?Id=1&ShowBy=DocType>).

Section 4 – Data Files

There are ten data files that make up the CIREN public data release, which is provided in a SAS-readable format (SAS Institute, Cary, NC). Six of the data files are based on the NASS-CDS schema and are derived using the same variable names, definitions, and attributes as described in Section 3.

The NASS-CDS-based files include the Acc_desc, Event, GV, VE, VI, and OA files. The Acc_desc file contains a text narrative describing the crash scene, sequence of events, and vehicle dynamics for each CIREN case. The Event file contains general information on vehicle class, general vehicle damage locations, and objects contacted by the vehicles in the crash. The GV (general vehicle) file contains detailed data on the vehicles involved in the crash, scene/environment data, crash energy, driver data, and rollover data. The VE (vehicle exterior) file contains data on the exterior vehicle damage, direction of force, and damage measurements. The VI (vehicle interior) file contains data related to damage and assessment of the vehicle interior, such as intrusions, glazing status, and steering assembly data. The OA (occupant assessment) file contains data on all occupants of the CIREN case vehicle. The OA file contains some occupant records that are not further detailed in the CIREN files due to those occupants not being enrolled as a CIREN occupant (e.g. occupant did not meet CIREN inclusion criteria or did not wish to participate in the program).

It should be noted that not all variables documented in the NASS-CDS Analytical User's Manual will be available in the CIREN data sets. As previously mentioned, CIREN cases are based on individual occupants involved in passenger vehicle crashes and are not part of a national sampling data system. Thus, certain variables from the NASS-CDS schema are not pertinent to the CIREN case. Please refer to Appendix B to view a comparison of variable inclusion in the CIREN and NASS-CDS public SAS data sets.

The unique CIREN files consist of the Case, Injury, Scenario and Injuryanalysis files. The Case file contains a wide array of specific data on the CIREN case occupants. The Injury file contains data describing the types, severity, and anatomic locations of injuries sustained by the CIREN case occupants. The Scenario file contains data relevant to the injury causations (BioTab) coded for the CIREN case occupants. The Injuryanalysis file contains a text narrative describing the occupant kinematics, injuries sustained, and injury causations for the CIREN case occupants.

Section 5 – Data Linkage

Since NASS-CDS data are captured for an entire crash and CIREN data are captured for an individual occupant, data users will need to utilize some new common key variables to appropriately join the NASS-CDS-based tables to the CIREN tables.

NASS-CDS annual data releases are provided in the SAS format and the CIREN public data release is assembled in the same format. To generate a data extract for a particular occupant from the NASS-CDS data sets, a researcher would link using the PSU number (PSU), case sequence number (CASENO), vehicle number (VEHNO), and occupant number (OCCNO) from a set of files specific to the year of the crash. Since CIREN cases are based on an individual occupant as opposed to a complete crash, the data are linked in a slightly different manner. Data files related to the crash and vehicle are linked by CASENO and/or VEHNO, whereas the OA file (based on NASS-CDS schema) requires CASENO, VEHNO and OCCNO for proper linkage. The unique CIREN data files (Case, Injury, Scenario, and Injuryanalysis) require different key variables for proper linkage. CIREN key variables include CIREN case unique identifier (CIRENID), unique injury identifier (INJURYID), and injury causation scenario identifier (SCENARIOID). These additional key variables will allow users to link files for identification of CIREN occupants to the injuries they sustained and the injury causation findings recorded shown below in Table 1.

Table 1. CIREN Data Key Variable Required for Proper Linkage.

SAS Data File	Record Specific	Key Variable(s)	Comments
Case	Occupant	CIRENID	
Injury	Injury	CIRENID, INJURYID	
Scenario	Injury Scenario	CIRENID, INJURYID, SCENARIOID	
Injuryanalysis	Occupant	CIRENID	LINENO is sequential per unique CIRENID to properly assemble the injury analysis text
Event	Crash Event	CASENO, VEHNO	
GV	Vehicle	CASENO, VEHNO	
VE	Vehicle	CASENO, VEHNO	
VI	Vehicle	CASENO, VEHNO	
OA	Occupant	CASENO, VEHNO, OCCNO	
Acc_desc	Crash	CASENO	LINENO is sequential per unique CASENO to properly assemble the crash description text

Section 6 – Case Files

6.1 CASE file

<u>Variable</u>	<u>SAS File Position</u>	<u>Short Description</u>
Identifiers =====		
CIRENID	1	CIREN case (individual subject person) unique identifier
CASEYEAR	22	calendar year when crash occurred
CASENO	2	NASSMAIN crash case identifier
VEHNO	132	NASSMAIN vehicle identifier for CIREN subject's vehicle
OCCNO	133	NASSMAIN occupant identifier for CIREN subject
Demographic/Anthropometric =====		
AGE_MNTH	23	age in months
AGE_YR	134	age in years
SEX	24	sex, including pregnancy status for females
WEIGHT	25	weight in kilograms
STATURE	26	standing height in centimeters
SEATED_HGT	29	seated height in centimeters
B_TO_K_LNTH	30	buttock-to-knee length in centimeters
KNEE_HGT	31	knee height, seated, in centimeters
RACE_ETHN	32	"race/ethnicity" as a single concept (2005 - 2006 only)
RACE	33	race (2007 - 2011)
ETHN	34	ethnicity (2007 - 2011)
EDUC	35	highest level of education
ALCH	36	alcohol use
SMOKE	37	tobacco use
MARITAL	38	marital status
LV_ARR	39	living arrangements
Injury Overview =====		
TOT_ICU	135	total days spent in the Intensive Care Unit (ICU)
TOT_VENT	136	total days with mechanical respiratory support
FATAL	27	fatality status
MAIS	69	maximum single Abbreviated Injury Scale (AIS) score
ISS	137	Injury Severity Score (ISS)
AIS_HEAD	40	highest AIS score in AIS body region "head"
AIS_FACE	41	highest AIS score in AIS body region "face"
AIS_NECK	42	highest AIS score in AIS body region "neck"
AIS_THRX	43	highest AIS score in AIS body region "thorax"
AIS ABD	44	highest AIS score in AIS body region "abdomen"
AIS_SPN	45	highest AIS score in AIS body region "spinal column"

AIS_U_EXT	46	highest AIS score in AIS body region “upper extremity”
AIS_L_EXT	47	highest AIS score in AIS body region “lower extremity”
AIS_EXTRNL	48	highest AIS score in AIS body region “external”
BT_HEAD	49	highest AIS score in BioTab body region “head”
BT_NECK	50	highest AIS score in BioTab body region “neck”
BT_C_SPN	51	highest AIS score in BioTab body region “cervical spine”
BT_SHLD	52	highest AIS score in BioTab body region “shoulder”
BT_U_ARM	53	highest AIS score in BioTab body region “upper arm”
BT_ELBOW	54	highest AIS score in BioTab body region “elbow”
BT_F_ARM	55	highest AIS score in BioTab body region “forearm”
BT_WRST	56	highest AIS score in BioTab body region “wrist”
BT_HAND	57	highest AIS score in BioTab body region “hand”
BT_THRX	58	highest AIS score in BioTab body region “thorax”
BT_T_SPN	59	highest AIS score in BioTab body region “thoracic spine”
BT_ABD	60	highest AIS score in BioTab body region “abdomen”
BT_L_SPN	61	highest AIS score in BioTab body region “lumbar spine”
BT_PLVS	62	highest AIS score in BioTab body region “pelvis”
BT_HIP	63	highest AIS score in BioTab body region “hip”
BT_THGH	64	highest AIS score in BioTab body region “thigh”
BT_KNEE	65	highest AIS score in BioTab body region “knee”
BT_LEG	66	highest AIS score in BioTab body region “lower leg”
BT_ANKL	67	highest AIS score in BioTab body region “ankle”
BT FOOT	68	highest AIS score in BioTab body region “foot”

Vital Signs/Glasgow Coma Scale (GCS) / Arterial Blood Gas (ABG)

=====

ADM_LAG	3	elapsed time from crash to first hospital admission
V1_LAG	4	elapsed time from crash to earliest pre-hospital vital signs
V1_SYST	115	earliest pre-hospital systolic blood pressure
V1_DIAS	116	earliest pre-hospital diastolic blood pressure
V1_PULSE	117	earliest pre-hospital pulse
V1_RESP	118	earliest pre-hospital respiration rate
V2_LAG	5	elapsed time from crash to first hospital vital signs
V2_SYST	119	first hospital systolic blood pressure
V2_DIAS	120	first hospital diastolic blood pressure
V2_PULSE	121	first hospital pulse
V2_RESP	122	first hospital respiration rate
G1_LAG	6	elapsed time from crash to earliest pre-hospital GCS
G1_GCS	123	earliest pre-hospital GCS
G1_STATE	21	earliest pre-hospital GCS state
G2_LAG	7	elapsed time from crash to first hospital GCS

G2_GCS	124	first hospital GCS
G2_STATE	22	first hospital GCS state
ABG_LAG	8	elapsed time from crash to first ABG measure
ABG_PH	9	arterial blood potential hydrogen (pH)
ABG_PO2	125	arterial blood oxygen partial pressure
ABG_PCO2	126	arterial blood carbon dioxide partial pressure
ABG_BASE	127	arterial blood base excess/deficit
ABG_HCO3	128	arterial blood bicarbonate

Death Circumstances

=====

D_LAG	10	elapsed time from crash to death
DEATH_LOC	28	location where the subject expired
CAUSE1	11	first cause of death
CAUSE2	14	second cause of death
CAUSE3	17	third cause of death
COMP1	12	first complication involved with death
COMP2	15	second complication involved with death
COMP3	18	third complication involved with death
COMOR1	129	first comorbidity involved with death
COMOR2	130	second comorbidity involved with death
COMOR3	131	third comorbidity involved with death
A_FCT1	13	first additional factor associated with death
A_FCT2	16	second additional factor associated with death
A_FCT3	19	third additional factor associated with death

Air Bag Availability/Function

=====

STRWHL_A	70	air bag availability/function, steering wheel
TOPIP_A	71	air bag availability/function, top instrument panel
MIDIP_A	72	air bag availability/function, mid instrument panel
BOTIP11_A	73	air bag availability/function, bottom instrument panel, left
BOTIP13_A	74	air bag availability/function, bottom instrument panel, right
SBKOUT11_A	75	air bag availability/function, front seat, outboard, left
SBKOUT13_A	76	air bag availability/function, front seat, outboard, right
SBKOUT21_A	77	air bag availability/function, second seat, outboard, left
SBKOUT23_A	78	air bag availability/function, second seat, outboard, right
DOOR11_A	79	air bag availability/function, front seat, door panel, left
DOOR13_A	80	air bag availability/function, front seat, door panel, right

DOOR21_A	81	air bag availability/function, second seat, door panel, left
DOOR23_A	82	air bag availability/function, second seat, door panel, left
ROOF1_A	83	air bag availability/function, roof rail, left
ROOF3_A	84	air bag availability/function, roof rail, right
STRWHL_E	85	air bag deployment event, steering wheel
TOPIP_E	86	air bag deployment event, top instrument panel
MIDIP_E	87	air bag deployment event, mid instrument panel
BOTIP11_E	88	air bag deployment event, bottom instrument panel, left
BOTIP13_E	89	air bag deployment event, bottom instrument panel, right
SBKOUT11_E	90	air bag deployment event, front seat, outboard, left
SBKOUT13_E	91	air bag deployment event, front seat, outboard, right
SBKOUT21_E	92	air bag deployment event, second seat, outboard, left
SBKOUT23_E	93	air bag deployment event, second seat, outboard, right
DOOR11_E	94	air bag deployment event, front seat, door panel, left
DOOR13_E	95	air bag deployment event, front seat, door panel, right
DOOR21_E	96	air bag deployment event, second seat, door panel, left
DOOR23_E	97	air bag deployment event, second seat, door panel, right
ROOF1_E	98	air bag deployment event, roof rail, left
ROOF3_E	99	air bag deployment event, roof rail, right
STRWHL_F	100	indications of air bag malfunction, steering wheel
TOPIP_F	101	indications of air bag malfunction, top instrument panel
MIDIP_F	102	indications of air bag malfunction, mid instrument panel
BOTIP11_F	103	indications of air bag malfunction, bottom instrument panel, left
BOTIP13_F	104	indications of air bag malfunction, bottom instrument panel, right
SBKOUT11_F	105	indications of air bag malfunction, front seat, outboard, left
SBKOUT13_F	106	indications of air bag malfunction, front seat, outboard, right
SBKOUT21_F	107	indications of air bag malfunction, second seat, outboard, left
SBKOUT23_F	108	indications of air bag malfunction, second seat, outboard, right
DOOR11_F	109	indications of air bag malfunction, front seat, door panel, left
DOOR13_F	110	indications of air bag malfunction, front seat, door panel, right
DOOR21_F	111	indications of air bag malfunction, second seat, door panel, left
DOOR23_F	112	indications of air bag malfunction, second seat, door panel, right
ROOF1_F	113	indications of air bag malfunction, roof rail, left
ROOF3_F	114	indications of air bag malfunction, roof rail, right

CIRENID position #1

Short definition: Unique identifier for a CIREN case (individual subject person).

Code elements: six or nine digits

Source: assigned by system

Range: N/A

Code Attributes: N/A

Comments:

(none)

CASEYEAR position #22

Short definition: The calendar year in which the crash occurred.

Code elements: four digits

Source: official documents

Range: 2005 - 2014

Code Attributes: N/A

Comments:

(none)

CASENO position #2

Short definition: Identifier for NASSMAIN crash case data, for the CIREN subject's crash.

Code elements: nine digits

Source: assigned by system

Range: N/A

Code Attributes: N/A

Comments:

(none)

VEHNO position #132

Short definition: The vehicle number (ordinal identifier) for the CIREN subject's vehicle in the NASSMAIN crash case data.

Code elements: one digit

Source: assigned by system

Range: 1, 2

Code Attributes: N/A

Comments:

VEHNO combined with CASENO comprise the unique identifier for the CIREN subject's vehicle within the NASS crash data.

OCCNO position #133

Short definition: The NASSMAIN occupant number (ordinal identifier) for the CIREN subject within his or her vehicle.

Code elements: one digit

Source: assigned by system

Range: 1 - 6

Code Attributes: N/A

Comments:

OCCNO combined with CASENO and VEHNO comprise the unique identifier for the CIREN subject within the NASS crash data.

AGE_MNTH position #23

Short definition: The CIREN subject's age in months, truncated to the last whole month.

Code elements: four digits

Source: Investigator determined

Range: 1 - 1200, 9999

Code Attributes:

9999 - unknown

Comments:

Computed within the CIREN application by subtracting the subject's date of birth from the crash date.

AGE_YR position #134

Short definition: The CIREN subject's age in years, truncated to the last whole year.

Code elements: three digits

Source: derived

Range: 0 - 100, 999

Code Attributes:

999 - unknown

Comments:

The CIREN database contains the subject's age in months. AGE_YR is computed from the original month-based age data.

SEX position #24

Short definition: CIREN subject's sex, including current pregnancy status for females.

Code elements: one digit

Source: Investigator determined

Range: 1 - 6, 9

Code attributes:

- 1 - male
- 2 - female, not pregnant
- 3 - female, pregnant, first trimester
- 4 - female, pregnant, second trimester
- 5 - female, pregnant, third trimester
- 6 - female, pregnant, unknown trimester
- 9 - unknown

Comments:

(none)

WEIGHT position #25

Short definition: CIREN subject's weight in kilograms.

Code elements: three digits

Investigator determined

Range: 3 - 350, 999

Code Attributes:

- 3 - 350 - the subject's measured weight in kilograms
- 999 - weight unknown

Comments:

(none)

STATURE position #26

Short definition: The CIREN subject's standing height, in centimeters.

Code elements: three digits

Source: Investigator determined

Range: 50 - 250, 999

Code Attributes:

50 - 250 - the subject's measured standing height in centimeters
999 - height unknown

Comments:

(none)

SEATED_HGT position #29

Short definition: The CIREN subject's seated height (from the chair seat to the top of the head), in centimeters.

Code elements: three digits

Source: Investigator determined

Range: 10 - 150, 999

Code Attributes:

10 - 150 - the subject's measured seated height in centimeters

999 - seated height unknown

Comments:

(none)

B_TO_K_LNTH position #30

Short definition: The CIREN subject's buttock-to-knee length, in centimeters.

Code elements: three digits

Source: Investigator determined

Range: 10 - 150, 999

Code Attributes:

10 - 150 - the subject's measured buttock-to-knee length in centimeters

999 - buttock-to-knee length unknown

Comments:

(none)

KNEE_HGT position #31

Short definition: The CIREN subject's knee height when seated ("knee height, seated"), in centimeters.

Code elements: three digits

Source: Investigator determined

Range: 10 - 150, 999

Code Attributes:

10 - 150 - the subject's measured knee height in centimeters

999 - knee height unknown

Comments:

(none)

RACE_ETHN position #32

Short definition: The CIREN subject's race/ethnicity (2005-2006)

Code elements: one digit

Source: Investigator determined

Range: 1 - 7, 9, null

Code Attributes:

- 1 - White (non-Hispanic)
- 2 - Black (non-Hispanic)
- 3 - White (Hispanic)
- 4 - Black (Hispanic)
- 5 - American Indian, Eskimo or Aleut
- 6 - Asian or Pacific Islander
- 7 - other
- 9 - unknown
- null - not measured (crash date 2007 - 2011)

Comments:

The single concept "race/ethnicity" was used until the end of 2006. This was split into two separate concepts, "race" and "ethnicity", beginning 01/01/2007.

RACE position #33

Short definition: The CIREN subject's race (2007-2011).

Code elements: one digit

Source: Investigator determined

Range: 1 - 5, 7, 9, null

Code Attributes:

- 1 - White
- 2 - Black
- 3 - Asian
- 4 - Native Hawaiian or other Pacific Islander
- 5 - American Indian or Alaska Native
- 7 - other
- 9 - unknown
- null - not measured (crash date 2005 - 2006)

Comments:

The single concept "race/ethnicity" was used until the end of 2006. This was split into two separate concepts, "race" and "ethnicity", beginning 01/01/2007.

ETHN position #34

Short definition: The CIREN subject's ethnicity (2007-2011).

Code elements: one digit

Source: Investigator determined

Range: 1, 2, 9, null

Code Attributes:

- 1 - Hispanic or Latino
- 2 - Not Hispanic or Latino
- 9 - unknown
- null - not measured (crash date 2005 - 2006)

Comments:

The single concept "race/ethnicity" was used until the end of 2006. This was split into two separate concepts, "race" and "ethnicity", beginning 01/01/2007.

EDUC position #35

Short definition: The CIREN subject's self-reported highest level of education.

Code elements: two digits

Source: Investigator determined

Range: 1 - 10, 99

Code Attributes:

- 1 - no schooling
- 2 - Preschool
- 3 - K thru 4
- 4 - 5 thru 8
- 5 - 9 thru 12
- 6 - HS or GED
- 7 - Vocational training
- 8 - Some college
- 9 - College degree
- 10 - Post Graduate training
- 99 - unknown

Comments:

(none)

ALCH position #36

Short definition: The CIREN subject's self-reported current alcohol use.

Code elements: one digit

Source: Investigator determined

Range: 1 - 5, 9

Code Attributes:

- 1 - don't drink
- 2 - occasional
- 3 - moderate
- 4 - heavy
- 5 - problem
- 9 - unknown

Comments:

(none)

SMOKE position #37

Short definition: The CIREN subject's self-reported current tobacco use.

Code elements: one digit

Source: Investigator determined

Range: 1 - 4, 9

Code Attributes:

- 1 - don't smoke
- 2 - occasional
- 3 - one pack per day
- 4 - more than one pack per day
- 9 - unknown

Comments:

(none)

MARITAL position #38

Short Definition: The CIREN subject's self-reported current marital status.

Code elements: one digit

Source: Investigator determined

Range: 1 - 6, 9

Code Attributes:

- 1 - never married
- 2 - married, living together
- 3 - married, not living together
- 4 - separated
- 5 - divorced and not remarried
- 6 - widowed and not remarried
- 9 - unknown

Comments:

(none)

LV_ ARR position #39

Short definition: The CIREN subject's self-reported current living arrangements.

Code elements: one digit

Source: Investigator determined

Range: 1 - 7, 9

Code Attributes:

- 1 - alone
- 2 - with spouse or unrelated partner
- 3 - with children
- 4 - with parent or guardian
- 5 - with relatives (other than spouse, parents or children)
- 6 - with nonrelatives
- 7 - in foster care
- 9 - unknown

Comments:

(none)

TOT_ICU position #135

Short definition: Total number of days spent by the subject in the Intensive Care Unit

Code elements: one, two or four digits

Source: Investigator determined

Range: -9999, 0 - 66

Code Attributes:

-9999 - total ICU days not known

x - xx - total days spent in the ICU (not necessarily continuous)

Comments:

(none)

TOT_VENT position #136

Short definition: Total number of days spent by the CIREN subject with mechanical respiratory assistance.

Code elements: one, two or four digits

Source: Investigator determined

Range: -9999, 0 - 66

Code Attributes:

-9999 - total ventilation days not known

x - xx - total ventilation days (not necessarily continuous)

Comments:

(none)

FATAL position #27

Short definition: The CIREN subject's fatality status

Code elements: one digit

Source: Investigator determined

Range: 1, 2 or 3

Code Attributes:

- 1 - not fatal
- 2 - fatal due to crash injuries
- 3 - fatal, ruled disease

Comments:

If the CIREN subject did not die within 30 days of the crash, that person is "not fatal".

If the CIREN subject died within 30 days of the crash as a result of crash injuries, that person is "fatal due to crash injuries".

If the CIREN subject died within 30 days of the crash as a result of disease rather than crash injuries, that person is "fatal, ruled disease".

MAIS position #69

Short definition: The CIREN subject's highest single AIS score.

Code elements: one digit

Source: derived

Range: 1 - 6

Code Attributes:

- 1 - minor injury
- 2 - moderate injury
- 3 - serious injury
- 4 - severe injury
- 5 - critical injury
- 6 - maximum (untreatable)

Comments:

See AISVERSION in the Injury file to determine AIS version used to generate MAIS.

ISS position #137

Short definition: The CIREN subject's Injury Severity Score (ISS).

Code elements: two digits

Source: derived

Range: 1 - 75, 99

Code Attributes: N/A

Comments:

Calculated from the CIREN subject's entire list of AIS injuries, per the ISS method.

See AISVERSION in the injury file to determine AIS version used to generate ISS.

AIS_HEAD position #40
AIS_FACE position #41
AIS_NECK position #42
AIS_THRX position #43 (thorax)
AIS_ABD position #44 (abdomen)
AIS_SPN position #45 (spine)
AIS_U-EXT position #46 (upper extremity)
AIS_L-EXT position #47 (lower extremity)
AIS_EXTRNL position #48 (external)

Short definition: The CIREN subject's maximum AIS scale score for each of the nine AIS body regions, respectively.

Code elements: one digit

Source: derived

Range: 0 - 6

Code Attributes:

- 0 - no injury in this body region
- 1 - minor injury
- 2 - moderate injury
- 3 - serious injury
- 4 - severe injury
- 5 - critical injury
- 6 - maximum (untreatable)

Comments:

(none)

BT_HEAD	position #49 (face/head)	BT_T_SPN	position #59 (thoracic spine)
BT_NECK	position #50	BT_ABD	position #60 (abdomen)
BT_C_SPN	position #51 (cervical spine)	BT_L_SPN	position #61 (lumbar spine)
BT_SHLD	position #52 (shoulder)	BT_PLVS	position #62 (pelvis)
BT_U_ARM	position #53 (upper arm)	BT_HIP	position #63
BT_ELLOW	position #54	BT_THGH	position #64 (thigh)
BT_F_ARM	position #55 (forearm)	BT_KNEE	position #65
BT_WRST	position #56 (wrist)	BT_LEG	position #66
BT_HAND	position #57	BT_ANKL	position #67 (ankle)
BT_THRX	position #58 (thorax)	BT FOOT	position #68

Short definition: The CIREN subject's maximum AIS scale score for each of the 20 BioTab body regions, respectively.

Code elements: one digit

Source: derived

Range: 0 - 6

Code Attributes:

- 0 - no BioTab record for this body region (see Comments, below)
- 1 - minor injury
- 2 - moderate injury
- 3 - serious injury
- 4 - severe injury
- 5 - critical injury
- 6 - maximum (untreatable)

Comments:

The BioTab protocol was incorporated into the CIREN program in phases. In the early phase (2005 - 2007), AIS-1 and some AIS-2 injuries were not eligible for BioTab coding. Accordingly, there are no BioTab records for certain injuries in these early cases to enable the derivation of "maximum AIS scale score for each of the BioTab Body Regions, respectively". Thus, for cases with crash date 2005 - 2007, code 0 (zero) means "no BioTab record" for AIS-1 and some AIS-2 injuries. For cases with crash date 2008 and newer, code 0 (zero) means "no injury in this BioTab body region".

ADM_LAG position #3

Short Definition: The elapsed time, in minutes, from the police-reported date/time of the crash to the subject's first admission to any medical facility.

Code elements: one to four digits

Source: calculated from field data based on official documents

Range: -9999, 0 - 840

Code attributes:

x - xxxx - elapsed time in minutes

-9999 - elapsed time unknown

Comments:

If either crash date/time or admission date/time at the first hospital is unknown, then ADM_LAG is unknown.

V1_LAG position #4

Short definition: The elapsed time, in minutes, from the police-reported date/time of the crash to the earliest time when vital signs were measured prior to admission to any health care facility.

Code elements: one to four digits

Source: calculated from field data based on official documents

Range: -9999, 1 - 550

Code Attributes:

x - xxxx - elapsed time in minutes

-9999 - elapsed time unknown

Comments:

The set of “V1” vital signs data presents measured vital signs for the earliest recorded measure prior to admission to any facility. This was operationalized in terms of the lag time, based on the date/time recorded for vital signs measurement: if the vital signs lag was less than the admissions lag (see ADM_LAG, above), the earliest such measure is presented.

V1_SYST position #115

Short Definition: Systolic blood pressure as recorded from the earliest measure of the subject's vital signs prior to the subject's first admission to any medical facility.

Code elements: one to four digits

Source: official documents

Range: -9999, -8888, 0 - 300

Code attributes:

0 - 300 - measured systolic blood pressure, mmHg

-8888 - palpable

-9999 - systolic blood pressure unknown

Comments:

(none)

V1_DIAS position #116

Short Definition: Diastolic blood pressure as recorded from the earliest measure of the subject's vital signs prior to the subject's first admission to any medical facility.

Code elements: one to four digits

Source: official documents

Range: -9999, -8888, 0 - 300

Code attributes:

0 - 300 - measured diastolic blood pressure, mmHg

-8888 - palpable

-9999 - diastolic blood pressure unknown

Comments:

(none)

V1_PULSE position #117

Short Definition: Pulse rate, in beats per minutes, as recorded from the earliest measure of the subject's vital signs prior to the subject's first admission to any medical facility.

Code elements: one to four digits

Source: official documents

Range: -9999, -8888, 0 - 300

Code attributes:

0 - 300 - measured pulse, beats per minute

-8888 - palpable

-9999 - pulse rate unknown

Comments:

(none)

V1_RESP position #118

Short Definition: Respiration rate, in respirations per minute, as recorded from the earliest measure of the subject's vital signs prior to the subject's admission to any medical facility.

Code elements: one to four digits

Source: official documents

Range: -9999, -8888, 0 - 110

Code attributes:

0 - 110 - measured respiration rate, respirations per minute

-8888 - agonal

-9999 - respiration rate unknown

Comments:

(none)

V2_LAG position #5

Short Definition: Elapsed time from the crash to the first measure of the subject's vital signs after admission to any medical facility, in minutes.

Code elements: one to four digits

Source: calculated from field data based on official documents

Range: -9999, 5 - 1145

Code attributes:

x - xxxx - minutes

-9999 - lag time unknown

Comments:

The set of "V2" vital signs data presents measured vital signs for the first recorded measure after admission to any facility. This was operationalized in terms of the lag time, based on the date/time recorded for vital signs measurement: if the vital signs lag was equal to or greater than the admissions lag (see ADM_LAG, above), the earliest such measure is presented.

V2_SYST position #119

Short Definition: Systolic blood pressure as recorded from the first measure of the subject's vital signs after admission to any medical facility.

Code elements: one to four digits

Source: official documents

Range: -9999, -8888, 0 - 300

Code attributes:

0 - 300 - measured systolic blood pressure, mmHg

-8888 - palpable

-9999 - systolic blood pressure unknown

Comments:

(none)

V2_DIAS position #120

Short Definition: Diastolic blood pressure as recorded from the first measure of the subject's vital signs after admission to any medical facility.

Code elements: one to four digits

Source: official documents

Range: -9999, -8888, 0 - 300

Code attributes:

0 - 300- measured diastolic blood pressure, mmHg

-8888 - palpable

-9999 - diastolic blood pressure unknown

Comments:

(none)

V2_PULSE position #121

Short Definition: Pulse rate as recorded from the first measure of the subject's vital signs after admission to any medical facility.

Code elements: one to four digits

Source: official documents

Range: -9999, -8888, 0 - 300

Code attributes:

0 - 300 - measured pulse, beats per minute

-8888 - palpable

-9999 - pulse rate unknown

Comments:

(none)

V2_RESP position #122

Short Definition: Respiration rate as recorded from the first measure of the subject's vital signs after admission to any medical facility.

Code elements: one to four digits

Source: official documents

Range: -9999, -8888, 0 - 110

Code attributes:

0 - 110 - measured respiration rate, respirations per minute

-8888 - agonal

-9999 - respiration rate unknown

Comments:

(none)

G1_LAG position #6

Short Definition: Elapsed time in minutes from the police-reported date/time of the crash to the earliest record of Glasgow Coma Scale (GCS) prior to the subject's admission to any medical facility.

Code elements: one to four digits

Source: calculated from field data based on official documents

Range: -9999, 1 - 547

Code attributes:

x - xxxx - minutes

-9999 - lag time unknown

Comments:

The set of "G1" GCS data presents measured GCS for the earliest recorded measure prior to admission to any facility. This was operationalized in terms of the lag time, based on the date/time recorded for GCS measurement: if the GCS lag was less than the admissions lag (see ADM_LAG, above), the earliest such measure is presented.

G1_GCS position #123

Short Definition: GCS score as recorded from the earliest assessment of the subject's level of consciousness prior to the subject's first admission to any medical facility.

Code elements: one or two digits

Source: official documents

Range: -99, 3 - 15

Code attributes:

3 - 15 - measured GCS

-99 - GCS unknown

Comments:

(none)

G1_STATE position #21

Short Definition: A measure of the subject's condition with respect to medical interventions or spinal cord injury when the earliest GCS score was recorded prior to the subject's first admission to any medical facility.

Code elements: one or two letters

Source: official documents

Range: L, T, S, TP, SC, U

Code attributes:

- L - legitimate (no interventions)
- T - intubated
- S - chemically sedated
- TP - intubated and chemically paralyzed
- SC - spinal cord injury
- U - unknown (not available; not recorded)

Comments:

(none)

G2_LAG position #7

Short Definition: Elapsed time, in minutes, from the police-reported date/time of the crash to the first record of GCS after admission to any medical facility.

Code elements: one to four digits

Source: calculated from field data based on official documents

Range: -9999, 6 - 1440

Code attributes:

x - xxxx - minutes

-9999 - lag time unknown

Comments:

The set of “G2” GCS data presents measured GCS for the first recorded measure following admission to any facility. This was operationalized in terms of the lag time, based on the date/time recorded for GCS measurement: if the GCS lag was equal to or greater than the admission lag (see ADM_LAG, above), the earliest such measure is presented.

G2_GCS position #124

Short Definition: GCS score as recorded from the first assessment of the subject's level of consciousness after admission to any medical facility.

Code elements: one or two digits

Source: official documents

Range: -99, 3 - 15

Code attributes:

3 - 15 - measured GCS

-99 - GCS unknown

Comments:

(none)

G2_STATE position #22

Short Definition: A measure of the subject's condition with respect to medical interventions or spinal cord injury when the first GCS score was recorded after admission to any medical facility.

Code elements: one or two letters

Source: official documents

Range: L, T, S, TP, SC, U

Code attributes:

- L - legitimate (no interventions)
- T - intubated
- S - chemically sedated
- TP - intubated and chemically paralyzed
- SC - spinal cord injury
- U - unknown (not available; not recorded)

Comments:

(none)

ABG_LAG position #8

Short definition: The elapsed time, in minutes, from the police-reported date/time of the crash to the first measure of the subject's arterial blood.

Code elements: one to four digits

Source: calculated from field data based on official documents

Range: -9999, 32 - 4320

Code attributes:

x - xxxx - lag time in minutes

-9999 - ABG lag time unknown

Comments:

The set of ABG data presents measured ABGs for the earliest recorded measure.

ABG_PH position #9

Short definition: Potential hydrogen (pH) as recorded from the first measure of the subject's arterial blood.

Code elements: one to four digits in the form x.xx

Source: official documents

Range: -9999, 7.0 - 7.59

Code attributes:

x - x.xx - measured pH

-9999 - pH unknown

Comments:

(none)

ABG_PO2 position #125

Short definition: Oxygen partial pressure as recorded from the first measure of the subject's arterial blood.

Code elements: one to four digits

Source: official documents

Range: -9999, 9 - 571

Code attributes:

x - xxx - measured PO2

-9999 - PO2 unknown

Comments:

(none)

ABG_PCO2 position #126

Short definition: Carbon dioxide partial pressure as recorded from the first measure of the subject's arterial blood.

Code elements: one to four digits

Source: official documents

Range: -9999, 3 - 200

Code attributes:

x - xxx - measured PCO2

-9999 - PCO2 unknown

Comments:

(none)

ABG_BASE position #127

Short definition: Base excess/deficit as recorded from the first measure of the subject's arterial blood.

Code elements: one to four digits

Source: official documents

Range: -9999, -16 - +89

Code attributes:

-xx - +xx - measured base excess/deficit

-9999 - base excess/deficit unknown

Comments:

(none)

ABG_HCO3 position #128

Short definition: Bicarbonate concentration as recorded from the first measure of the subject's arterial blood.

Code elements: one to four digits

Source: official documents

Range: -9999, 2 - 47

Code attributes:

x - xxx - measured HCO3

-9999 - HCO3 unknown

Comments:

(none)

D_LAG position #10

Short Definition: For subjects who were “fatal” or “fatal, ruled disease” (see the discussion for the variable FATAL, above), elapsed time from the police-reported date/time of the crash to the officially reported date/time of death, expressed in days and hundredths of a day.

Code Elements: four digits

Source: official documents

Range: null, -9999, 0 - 25.71

Code attributes:

xx.xx - lag time in days (e.g., “1.25” = 1 and 1/4 days = 1 day and 6 hours = 30 hours)

null - not fatal

-9999 - fatal, time from crash to death unknown

Comment:

(none)

DEATH_LOC position #28

Short definition: The location where a crash victim died.

Code elements: one or two digits

Source: Investigator determined

Range: -1, 0 - 14

Code Attributes:

- 0 - not fatal
- 1 - at crash scene ("DOS" - dead on scene)
- 2 - emergency medical services (EMS) vehicle
- 3 - emergency department
- 4 - operating room
- 5 - radiology
- 6 - intensive care unit (ICU)
- 7 - intermediate care unit
- 8 - floor
- 9 - unknown
- 10 - pre-hospital, not specified
- 11 - other hospital
- 13 - resuscitation room - not in the Emergency Department

Comments:

(none)

CAUSE1 position #11 (first cause of death)
CAUSE2 position #14 (second cause of death)
CAUSE3 position #17 (third cause of death)

Short Definition: For subjects who were “fatal” or “fatal, ruled disease” (see the discussion for the variable FATAL, above), the injury or other pathology that was judged to be the cause of death.

Code elements: four or seven digits

Source: investigator determined based on official documents

Range: null, -2222, -8869, -8875, -8876, -9999, xxxxxxxx (AIS code)

Code attributes:

- null - not fatal
- xxxxxxx - AIS code for a specific injury
- 2222 - fatal, additional cause not cited (CAUSE2 and CAUSE3 only)
- 8869 - comorbidity
- 8875 - complication
- 8876 - non-injury
- 9999 - unknown

Comment:

(none)

COMP1 position #12 (first complication)
 COMP2 position #15 (second complication)
 COMP3 position #18 (third complication)

Short Definition: For subjects whose cause of death was judged to involve a complication (see CAUSE1, CAUSE2, CAUSE3), the indicated complication.

Code elements: four digits or null

Source: investigator determined, based on official documents

Range: null, -8873, -2222, x - x

Code attributes:

- null - not fatal
- 8873 - multiple complications
- 2222 - fatal, complication not cited
- 2901 - RENAL FAILURE
- 2902 - URETHRAL INJURY (IATROGENIC)
- 2903 - UTI
- 2904 - OTHER - RENAL
- 3001 - DISSEMINATED FUNGAL INFECTION
- 3002 - INTRA-ABDOMINAL ABSCESS
- 3003 - NECROTIZING FASCITIS
- 3004 - WOUND INFECTION (NON-SURGICAL)
- 3005 - WOUND INFECTION (SURGICAL)
- 3006 - BACTEREMIA
- 3007 - OTHER - INFECTION
- 3008 - WOUND INFECTION
- 3701 - COMPARTMENT SYNDROME
- 3702 - LOSS OF OPERATIVE REDUCTION/FIXATION
- 3703 - OSTEOMYELITIS
- 3704 - ORTHOPEDIC WOUND INFECTION
- 3705 - RHABDOMYOLYSIS
- 3706 - SKIN BREAKDOWN
- 3707 - OTHER - MUSCULAR/SKELETAL
- 3801 - ALCOHOL WITHDRAWAL
- 3802 - DIABETES INSIPIDUS
- 3803 - FAILED NON-OPERATIVE SDH/EDH
- 3804 - PROGRESSION OF ORIGINAL NEUROLOGIC INSULT
- 3805 - SIADH
- 3806 - VENTRICULITIS - POST OP (IATROGENIC)
- 3807 - NEUROPRAXIA (IATROGENIC)
- 3808 - OTHER - NEUROLOGIC
- 3901 - DVT (LOWER EXTREMITY)
- 3902 - GRAFT INFECTION
- 3903 - THROMBOSIS AND/OR EMBOLUS

COMP1 position #11 (first complication) continued

COMP2 position #15 (second complication)

COMP3 position #18 (third complication)

- 3904 - OTHER - VASCULAR
- 4101 - ANESTHETIC COMPLICATION
- 4102 - HYPOTHERMIA
- 4103 - UNEXPECTED READMISSION - HOSPITAL
- 4104 - UNEXPECTED POST- OPERATIVE HEMORRHAGE
- 4105 - MANAGED CARE ISSUE
- 4106 - UNEXPECTED RETURN TO OR
- 4107 - UNEXPECTED READMISSION - ICU
- 4108 - FAILED NON-OP MANAGEMENT - SPLEEN
- 4109 - FAILED NON-OP MANAGEMENT - LIVER
- 4110 - NON THERAPEUTIC LAPAROTOMY
- 4111 - OTHER - COMPLICATIONS
- 4201 - DEATH, PREVENTABLE
- 4202 - DEATH, NON-PREVENTABLE
- 4203 - DEATH, POTENTIALLY PREVENTABLE
- 4301 - DELAY IN TRAUMA TEAM ACTIVATION
- 4302 - DELAY TO OR
- 4303 - DELAY MD RESPONSE
- 4304 - DELAY IN OBTAINING SUBSPECIALTY CONSULT
- 4305 - DELAY IN DIAGNOSIS
- 4306 - ERROR IN DIAGNOSIS
- 4307 - ERROR IN JUDGEMENT
- 4308 - ERROR IN TECHNIQUE
- 4309 - INCOMPLETE HOSPITAL RECORD
- 4310 - DELAY IN OBTAINING TRAUMA CONSULT
- 4311 - DELAY IN PRESENTATION (ACCEPTABLE)
- 4312 - NURSING ISSUE
- 4313 - OTHER - ERRORS/DELAYS
- 4401 - OTHER - PREHOSPITAL AIRWAY
- 4402 - FLUIDS
- 4403 - SPINAL IMMOBILIZATION
- 4404 - TRIAGE
- 4405 - OTHER - PREHOSPITAL
- 4701 - ESOPHAGEAL INTUBATION
- 4702 - EXTUBATION - UNINTENTIONAL
- 4703 - OTHER - HOSPITAL AIRWAY
- 4704 - MAINSTEM INTUBATION
- 4801 - ARDS
- 4802 - ASPIRATION / PNEUMONIA
- 4803 - EMPYEMA
- 4804 - FAT EMBOLUS
- 4805 - HEMOTHORAX (RETAINED)
- 4806 - PNEUMONIA
- 4807 - PNEUMOTHORAX (BAROTRAUMA)
- 4808 - PNEUMOTHORAX (IATROGENIC)
- 4809 - PNEUMOTHORAX (RECURRENT)
- 4810 - PNEUMOTHORAX (TENSION)
- 4811 - PULMONARY EMBOLUS

COMP1 position #12 (first complication)
COMP2 position #15 (second complication)
COMP3 position #18 (third complication)

continued

4812 - OTHER - PULMONARY
4813 - PNEUMOTHORAX
4901 - CARDIAC ARREST (UNEXPECTED)
4902 - MI
4903 - OTHER - CARDIOVASCULAR
5001 - ANASTOMOTIC LEAK
5002 - DEHISCENCE /+ EVISCERATION
5003 - FISTULA (OTHER THAN PANCREATIC)
5004 - HEMORRHAGE - GI
5005 - BOWEL INJURY (IATROGENIC)
5006 - OTHER - GI
5101 - ACALCULOUS CHOLECYSTITIS
5102 - LIVER FAILURE (JAUNDICE)
5103 - PANCREATIC FISTULA
5104 - PANCREATITIS
5105 - SPLENIC INJURY (IATROGENIC)
5106 - OTHER - HEPATIC
5201 - COAGULOPATHY
5202 - TRANSFUSION COMPLICATION
5203 - OTHER - HEMATOLOGIC
5901 - NONE

Comment:

(none)

COMOR1 position #129 (first comorbidity)
 COMOR2 position #130 (second comorbidity)
 COMOR3 position #131 (third comorbidity)

Short definition: For subjects whose cause of death was judged to involve a co-morbid condition (see CAUSE1, CAUSE2, CAUSE3), the indicated comorbidity.

Code elements: three or four digits

Source: investigator determined, based on official documents

Range: null, -9999, -9997, -2222, 101 - 1601

Code attributes:

- null - not fatal
- 9999 - OTHER * UNKNOWN
- 9997 - OTHER * Specify [specify text not included]
- 2222 - fatal, comorbidity not cited

- 101 - GENERAL * RECENT DELIVERY
- 102 - GENERAL * PREGNANCY
- 301 - CARDIAC DISEASE * HISTORY OF CARDIAC SURGERY
- 302 - CARDIAC DISEASE * CORONARY ARTERY DISEASE
- 303 - CARDIAC DISEASE * CONGESTIVE HEART FAILURE
- 304 - CARDIAC DISEASE * COR PULMONALE
- 305 - CARDIAC DISEASE * MYOCARDIAL INFARCTION
- 306 - CARDIAC DISEASE * HYPERTENSION
- 401 - DIABETES * INSULIN DEPENDENT
- 402 - DIABETES * NON-INSULIN DEPENDENT (ON ORAL MEDICATION)
- 501 - GASTROINTESTINAL DISEASE * PEPTIC ULCER DISEASE
- 502 - GASTROINTESTINAL DISEASE * GASTRIC OR ESOPHAGEAL VARICES
- 503 - GASTROINTESTINAL DISEASE * PANCREATITIS
- 504 - GASTROINTESTINAL DISEASE * INFLAMMATORY BOWEL DISEASE
- 601 - HEMATOLOGIC DISORDERS * ACQUIRED COAGULOPATHY
- 602 - HEMATOLOGIC DISORDERS * BLOOD THINNER THERAPY
- 603 - HEMATOLOGIC DISORDERS * HEMOPHILIA
- 604 - HEMATOLOGIC DISORDERS * PRE-EXISTING ANEMIA
- 701 - PSYCHIATRIC DISORDERS * HISTORY OF PSYCHIATRIC DISORDERS
- 801 - IMMUNOSUPPRESSED THERAPY * TRANSPLANTS
- 802 - IMMUNOSUPPRESSED THERAPY * HIV/AIDS
- 803 - IMMUNOSUPPRESSED THERAPY * ROUTINE STEROID USE
- 804 - IMMUNOSUPPRESSED THERAPY * ACTIVE CHEMOTHERAPY
- 901 - LIVER DISEASE * BILIRUBIN >2MG O/O (ON ADMISSION)
- 902 - LIVER DISEASE * DOCUMENTED HISTORY OF CIRRHOSIS
- 1001 - MALIGNANCY * UNDERGOING CURRENT THERAPY FOR MALIGNANCY
- 1002 - MALIGNANCY * CONCURRENT OR EXISTENCE OF METASTASIS
- 1101 - MUSCULOSKELETAL * OSTEOPOROSIS
- 1102 - MUSCULOSKELETAL * RHEUMATOID ARTHRITIS

COMOR1 position #129 (first comorbidity)
COMOR2 position #130 (second comorbidity)
COMOR3 position #131 (third comorbidity)

(continued)

1103 - MUSCULOSKELETAL * SYSTEMIC LUPUS ERYTHEMATOUS
1104 - MUSCULOSKELETAL * PRIOR JOINT REPLACEMENT
1105 - MUSCULOSKELETAL * PRIOR SURGICAL IMPLANTS
1201 - NEUROLOGIC * SPINAL CORD INJURY
1202 - NEUROLOGIC * MULTIPLE SCLEROSIS
1203 - NEUROLOGIC * ALZHEIMERS DISEASE
1204 - NEUROLOGIC * SEIZURES
1205 - NEUROLOGIC * CHRONIC DEMYELINATING DISEASE
1206 - NEUROLOGIC * CHRONIC DEMENTIA
1207 - NEUROLOGIC * ORGANIC BRAIN SYNDROME
1208 - NEUROLOGIC * PARKINSONS DISEASE
1209 - NEUROLOGIC * CVA/HEMIPARESIS (STROKE WITH RESIDUAL)
1301 - PULMONARY * DOCUMENTED PRIOR HX OF PULM DX W/ ONGOING ACTIVE TX
1302 - PULMONARY * ASTHMA
1303 - PULMONARY * CHRONIC OBSTRUCTIVE PULMONARY DISEASE
1304 - PULMONARY * CHRONIC PULMONARY CONDITION
1401 - RENAL DISEASE * SERUM CREATININE >2MG O/O (ON ADMISSION)
1402 - RENAL DISEASE * DIALYSIS (EXCLUDES TRANSPLANT PATIENTS)
1501 - SUBSTANCE ABUSE * CHRONIC DRUG USER
1502 - SUBSTANCE ABUSE * CHRONIC ALCOHOL USER
1601 - OBESITY * OBESITY

Comment:

(none)

A_FCT1 position #13 (first associated factor)
A_FCT2 position #16 (second associated factor)
A_FCT3 position #19 (third associated factor)

Short Definition: For subjects who were “fatal” or “fatal, ruled disease” (see the discussion for the variable FATAL, above), an additional injury or other factor that was judged to be associated with the cause of death.

Code elements: null, four or seven digits

Source: investigator determined

Range: null, -8874, -2222, xxxxxxxx (AIS code)

Code attributes:

- null - not fatal
- 8874 - multiple injuries
- 2222 - fatal, associated factor not cited
- xxxxxxx - AIS code for a specific injury that was judged to be an associated factor

Comment:

(none)

Air Bag Availability/Deployment Status (suffix “_A”)

STRWHL_A	position #70	(steering wheel)
TOPIP_A	position #71	(top instrument panel)
MIDIP_A	position #72	(middle instrument panel)
BOTIP11_A	position #73	(bottom instrument panel/knee bolster, front left)
BOTIP13_A	position #74	(bottom instrument panel/knee bolster, front right)
SBKOUT11_A	position #75	(seat back mounted, outboard side, front left)
SBKOUT13_A	position #76	(seat back mounted, outboard side, front right)
SBKOUT21_A	position #77	(seat back mounted, outboard side, second left)
SBKOUT23_A	position #78	(seat back mounted, outboard side, second right)
DOOR11_A	position #79	(door panel, front left)
DOOR13_A	position #80	(door panel, front right)
DOOR21_A	position #81	(door panel, second left)
DOOR23_A	position #82	(door panel, second right)
ROOF1_A	position #83	(roof side rail, left)
ROOF3_A	position #84	(roof side rail, right)

Short definition: Air bag availability/function for the indicated location.

Code elements: one digit

Source: derived from vehicle inspection data

Range: 0, 1, 2, 9

Code attributes

- 0 - air bag not available
- 1 - air bag available, not deployed
- 2 - air bag deployed
- 9 - unknown

Comments:

The attribute “unknown” means “unknown if air bag available/unknown if deployed.” The use of “unknown” for this variable is usually the result of vehicle components that were missing at the time of the vehicle inspection.

Air Bag Deployment Event Number (suffix “_E”)

STRWHL_E	position #85	(steering wheel)
TOPIP_E	position #86	(top instrument panel)
MIDIP_E	position #87	(middle instrument panel)
BOTIP11_E	position #88	(bottom instrument panel/knee bolster, front left)
BOTIP13_E	position #89	(bottom instrument panel/knee bolster, front right)
SBKOUT11_E	position #90	(seat back mounted, outboard side, front left)
SBKOUT13_E	position #91	(seat back mounted, outboard side, front right)
SBKOUT21_E	position #92	(seat back mounted, outboard side, second left)
SBKOUT23_E	position #93	(seat back mounted, outboard side, second right)
DOOR11_E	position #94	(door panel, front left)
DOOR13_E	position #95	(door panel, front right)
DOOR21_E	position #96	(door panel, second left)
DOOR23_E	position #97	(door panel, second right)
ROOF1_E	position #98	(roof side rail, left)
ROOF3_E	position #99	(roof side rail, right)

Short definition: NASS Event sequence number for the crash event that was judged to cause the air bag deployment for the indicated location.

Code elements: one or two digits

Source: derived from vehicle inspection data

Range: 0, 1 - 30, 98, 99

Code attributes:

- 0 - air bag not available
- 1 - 30 - NASS event sequence number for air bag deployment
- 98 - air bag available, not deployed
- 99 - unknown event or unknown if deployed

Comments:

(none)

Indications of Air Bag Malfunction (suffix “_F”)

STRWHL_F	position #100	(steering wheel)
TOPIP_F	position #101	(top instrument panel)
MIDIP_F	position #102	(middle instrument panel)
BOTIP11_F	position #103	(bottom instrument panel/knee bolster, front left)
BOTIP13_F	position #104	(bottom instrument panel/knee bolster, front right)
SBKOUT11_F	position #105	(seat back mounted, outboard side, front left)
SBKOUT13_F	position #106	(seat back mounted, outboard side, front right)
SBKOUT21_F	position #107	(seat back mounted, outboard side, second left)
SBKOUT23_F	position #108	(seat back mounted, outboard side, second right)
DOOR11_F	position #109	(door panel air bag, front left)
DOOR13_F	position #110	(door panel air bag, front right)
DOOR21_F	position #111	(door panel air bag, second left)
DOOR23_F	position #112	(door panel air bag, second right)
ROOF1_F	position #113	(roof side rail air bag, left)
ROOF3_F	position #114	(roof side rail air bag, right)

Short definition: The investigator’s assessment of possible air bag malfunction for the indicated location.

Code elements: one digit

Source: derived from vehicle inspection data

Range: 0, 1, 2, 9

Code attributes:

- 0 - air bag not available
- 1 - no indication of air bag malfunction
- 2 - yes, possible air bag malfunction
- 9 - unknown

Comments:

(none)

6.2 INJURY file

<u>Variable</u>	<u>SAS File Position</u>	<u>Short Description</u>
Identifiers =====		
CIRENID	1	CIREN case unique identifier (individual subject person)
CASEYEAR	27	calendar year when crash occurred
INJURYID	2	unique identifier for an individual injury
AIVERSION	28	version of AIS used to encode the injury
Injury =====		
AISCODE	3	AIS code
INJRANK	18	injury severity rank
P_ASPECT1	7	first primary aspect
P_ASPECT2	8	second primary aspect
S_ASPECT1	9	first secondary aspect
S_ASPECT2	10	second secondary aspect
S_ASPECT3	11	third secondary aspect
S_ASPECT4	12	fourth secondary aspect
S_ASPECT5	13	fifth first secondary aspect
T_ASPECT1	14	first tertiary aspect
T_ASPECT2	15	second tertiary aspect
ICDM1	4	first International Classification of Disease (ICDM) code
ICDM2	5	second ICDM code
ICDM3	6	third ICDM code
Injury Source Contacts and Intrusions =====		
CONT1	16	first contacted component
CONT2	17	second contacted component
INTR1_LOC	21	first intrusion location
INTR2_LOC	22	second intrusion location
INTR1_COMP	19	first intruding component
INTR2_COMP	20	second intruding component
INTR1_DIR	25	first intrusion direction
INTR2_DIR	26	second intrusion direction
INTR1_MAG	23	first intrusion magnitude
INTR2_MAG	24	second intrusion magnitude

CIRENID position #1

Short definition: Unique identifier for a CIREN case (individual subject person).

Code elements: six or nine digits

Source: assigned by system

Range: N/A

Code attributes: N/A

Comments:

(none)

CASEYEAR position #27

Short definition: The calendar year in which the crash occurred.

Code elements: four digits

Source: official documents

Range: 2005 - 2014

Code Attributes: N/A

Comments:

(none)

INJURYID position #2

Short definition: Identifier for each specific injury.

Code elements: nine digits

Source: assigned by system

Range: N/A

Code attributes: N/A

Comment:

CIRENID combined with INJURYID comprise the unique identifier for a specific injury within a CIREN case.

AISVERSION position #28

Short definition: Version of the AIS system used to encode this injury.

Code elements: four digits

Source: assigned by system

Range: 1998, 2005

Code attributes: 1998, 2005

Comments:

AIS-1998 was used by CIREN for cases with crash date prior to 05/31/2010. AIS-2005 Update 2008 was adopted as of 06/01/2010.

AISCODE position #3

Short definition: AIS code for each specific injury.

Code elements: seven digits

Source: investigator determined

Range: structured code scheme always including seven digits

Code attributes: see the relevant AAAM/NHTSA Injury Coding Manuals

Comments:

The Abbreviated Injury Scale is an injury coding system developed by the Association for the Advancement of Automotive Medicine (AAAM). AIS code definitions are available from the Association for the Advancement of Automotive Medicine (www.aaam.org)

INJRANK position #18

Short definition: This injury's rank with respect to threat to life; rank 1 is most threatening.

Code elements: one or two digits

Range: 1, 2, 3 ... 99

Source: Investigator determined

Code attributes:

- 1 - rank 1
- 2 - rank 2
- 3 - rank 3
- etc.
- 99 - not ranked

Comments:

Three most severe injuries with an injury severity of 2 and greater are required to be ranked by the CIREN medical investigator in regards to threat to life. If two or more injuries are judged to represent the same degree of threat, they can be assigned the same rank (that is, there may be more than one injury with the same rank). Teams have the ability to rank injuries beyond the third level.

P_ASPECT1 position #7 (first Primary Aspect)
P_ASPECT2 position #8 (second Primary Aspect)

Short Definition: An elaboration on the AIS code system that provides information about the location of each specific injury.

Code elements: one digit

Source: Investigator determined

Range: 0 - 9, null

Code attributes:

- null - no Primary Aspect
- 1 - right
- 2 - left
- 3 - bilateral
- 4 - central
- 5 - anterior
- 6 - posterior
- 7 - superior
- 8 - inferior
- 9 - multiple/unknown
- 0 - whole region

Comments:

Primary Aspect codes, devised by the NHTSA, provide supplemental information about the location of specific injuries. One and only one Primary Aspect code is entered for each AIS-2005 code and most AIS-1998 codes. Some CIREN injuries coded under AIS-98 have no Primary Aspect code entered; other CIREN AIS-98 injuries may have as many as five Primary Aspect codes. Two Primary Aspect codes are included in this database.

S_ASPECT1 position #9 (first Secondary Aspect)
 S_ASPECT2 position #10 (second Secondary Aspect)
 S_ASPECT3 position #11 (third Secondary Aspect)
 S_ASPECT4 position #12 (fourth Secondary Aspect)
 S_ASPECT5 position #13 (fifth Secondary Aspect)

Short definition: An elaboration of the AIS code system that provides further detail, beyond Primary Aspect, about the location of certain specific injuries.

Code Elements: two or three digits

Range: 10 - 914, null

Source: Investigator determined

Code attributes:

null - no Secondary Aspect

10 - Upper Arm	37 - T11T12
11 - Elbow	38 - L1L2
12 - Forearm	39 - L2L3
13 - Wrist	40 - L3L4
14 - Hand/Digits	41 - L4L5
15 - Buttock	43 - C1
16 - Thigh	44 - C2
17 - Knee	45 - C3
18 - Lower Leg	46 - C4
19 - Ankle	47 - C5
20 - Foot/Toes	48 - C6
21 - C1C2	49 - C7
22 - C2C3	50 - T1
23 - C3C4	51 - T2
24 - C4C5	52 - T3
25 - C5C6	53 - T4
26 - C6C7	54 - T5
27 - T1T2	55 - T6
28 - T2T3	56 - T7
29 - T3T4	57 - T8
30 - T4T5	58 - T9
31 - T5T6	59 - T10
32 - T6T7	60 - T11
33 - T7T8	61 - T12
34 - T8T9	62 - L1
35 - T9T10	63 - L2
36 - T10T11	64 - L3

Secondary Aspect Code attributes (continued)

65 - L4	292 - Whole Front
66 - L5	293 - Whole Back
68 - Cauda Equina	294 - Coccyx
110 - Lateral	295 - Sacrum
121 - medial	296 - Right Ilium
200 - Head/Face	297 - Left Ilium
201 - Cheek	298 - Right Ischium
202 - Chin	299 - Left Ischium
205 - Lips	300 - Right Pubic Ramus
206 - Ear	301 - Left Pubic Ramus
207 - Nose	302 - Pubic Symphysis
220 - Temporal	303 - Right Sacroiliac Joint
221 - Medulla	304 - Left Sacroiliac Joint
222 - Right Lobe	305 - Right Eyebrow
223 - Hypothalamus	306 - Left Eyebrow
224 - Parietal	307 - Right Forehead
225 - Midbrain	308 - Left Forehead
226 - Nasal Spine	309 - Eyelid
227 - Left Lobe	310 - Mid Forehead
228 - Pons	311 - Lateral Wall
230 - Occipital-Right	889 - R Rib 1
231 - Occipital-Left	890 - R Rib 2
232 - Anterior Fossa	891 - R Rib 3
233 - Middle Fossa	892 - R Rib 4
234 - Posterior Fossa	893 - R Rib 5
235 - Frontal-Right	894 - R Rib 6
236 - Frontal-Left	895 - R Rib 7
240 - Ventricle	896 - R Rib 8
241 - Atria	897 - R Rib 9
250 - Right Front	898 - R Rib 10
251 - Left Front	899 - R Rib 11
252 - Right Back	900 - R Rib 12
253 - Left Back	901 - L Rib 1
260 - C7T1	902 - L Rib 2
261 - T12L1	903 - L Rib 3
262 - L5S1	904 - L Rib 4
270 - Ascending	905 - L Rib 5
271 - Descending	906 - L Rib 6
280 - Upper Extremity	907 - L Rib 7
281 - Torso	908 - L Rib 8
282 - Flank	909 - L Rib 9
283 - Groin	910 - L Rib 10
284 - Lower Extremity	911 - L Rib 11
285 - OC1	912 - L Rib 12
286 - Arch	913 - Shoulder
290 - Central Front	914 - Hip
291 - Central Back	

Comments:

Some Secondary Aspect codes were introduced while AIS-1998 was in use, but they were not applied systematically. Secondary Aspect codes were formally adopted as a component of AIS-2005 as of 06/01/2010.

Not all AIS-2005 injury codes are eligible for a Secondary Aspect; other AIS-2005 injury codes may require as many as 24 Secondary Aspect codes. The first five Secondary Aspect codes are included in this database.

T_ASPECT1 position #14 (first tertiary aspect)
T_ASPECT2 position #15 (second tertiary aspect)

Short definition: An elaboration of the AIS code system that provides further detail, beyond Primary and Secondary Aspects, about the location of certain injuries.

Code elements: four digits

Range: 1000 - 1024, null

Source: Medical Investigator / Radiology

Code attributes:

 null - no Tertiary Aspect
 1000 - Right Process
 1001 - Left Process
 1010 - Stable
 1011 - Unstable
 1020 - Anterior Rib
 1021 - Lateral Rib
 1022 - Anterio-lateral Rib
 1023 - Posterio-lateral Rib
 1024 - Posterior Rib

Comments:

Some Tertiary aspects were introduced while AIS-1998 was in use, but they were not applied systematically. Tertiary Aspect codes were formally adopted as a component of AIS-2005 as of 06/01/2010. Tertiary aspect codes are restricted to spinal column and rib cage injuries.

- Right/Left Process – indicates the anatomical location for a vertebral transverse process fracture
- Stable/Unstable – indicates if the fracture is stable or unstable
- Anterior, Lateral, Anterio-lateral, Posterio-lateral and Posterior Rib indicates the location for the majority of the rib fractures

ICDM1 position #4 (first ICDM)
ICDM2 position #5 (second ICDM)
ICDM3 position #6 (third ICDM)

Short definition: International Classification of Diseases, Ninth Revision, Clinical Modification (ICD-9-CM)⁴

Code elements: three, four or five digits, most including a decimal point

Range: 800.0 - 999.9, null

Source: Investigator determined

Code attributes:

xxx, xxx.x, xxx.xx (refer to the ICD-9-CM manual)
null - no additional ICDM code

Comments:

There will always be at least one ICDM code. Some single AIS injury codes require more than one ICDM code. A few CIREN injuries have as many as seven ICDM codes entered; the first three are included in this database.

CONT1 position #16 (first contact point)

CONT2 position #17 (second contact point)

Short definition: A component on the subject's motor vehicle that was identified as a contact point and which is thought to be associated with the current injury.

Code elements: three digit code, or null

Range: null, 001 - 594

Source: Investigator determined

Code attributes:

null	no contact point linked
001	Windshield
002	Mirror
003	Sunvisor
004	Steering wheel rim
005	Steering wheel hub/spoke
006	Steering wheel (combination of codes 004 and 005)
007	Steering column,transmission selector lever,other attachment
008	Cellular telephone or CB radio
009	Add on equipment (e.g., tapedeck, air conditioner)
010	Left instrument panel and below
011	Center instrument panel and below
012	Right instrument panel and below
013	Glove compartment door
014	Knee bolster
015	(Driver only) Windshield including one or more of: front header, left A-pillar, left instrument panel, mirror, steering assembly
016	(Front Right Passenger only) Windshield including one or more of: front header, right A-pillar, right instrument panel, mirror
017	Windshield reinforced by exterior object, (specify)
019	Other front object (specify)
020	Sunvisor reinforced by front header
021	Left Instrument Panel
022	Center Instrument Panel
023	Right Instrument Panel
024	Left lower instrument panel (includes knee bolster)
025	Center lower instrument panel (includes knee bolster)
026	Right lower instrument panel (includes knee bolster)
051	Left side interior surface, excl. hardware or armrests
052	Left side hardware or armrest

CONT1 position #16 (first contact point)
CONT2 position #17 (second contact point)

(continued)

053 Left A (A 1/A2)-pillar
054 Left B-pillar
055 Other left pillar (specify)
056 Left side window glass
057 Left side window frame
058 Left side window sill
059 Lt side glass+frame,win sill,A pill,B pill,or roof side rail
060 Other left side object (specify)
061 Left side glass (Laminated) reinforced by ext obj (specify)
062 Left side panel forward of A1/A2 pillar
063 Left side panel rear of the B-pillar
101 Right side interior surface excl hardware or armrests
102 Right side hardware or armrest
103 Right A (A 1/A2)-pillar
104 Right B-pillar
105 Other right pillar (specify)
106 Right side window glass
107 Right side window frame
108 Right side window sill
109 Rt side glass+frame,win sill,A pill,B pill,or roof side rail
110 Other right side object (specify)
111 Right side glass (Laminated) reinforced by ext obj (specify)
112 Right side panel forward of A1/A2 pillar
113 Right side panel rear of the B-pillar
151 Seat, back support
152 Belt restraint webbing/buckle
153 Belt restraint b pillar or door frame attachment point
154 Other restraint system component (specify)
155 Head restraint system
160 Other occupants (specify)
161 Interior loose objects (specify)
162 Child safety seat (specify)
163 Other interior object (specify)
164 Center console first row
165 Center console second row
166 Center console other row
167 Fold down armrest first row
168 Fold down armrest second row
169 Fold down armrest other row

CONT1 position #16 (first contact point)

(continued)

CONT2 position #17 (second contact point)

- 170 Air bag-driver side
- 175 Air bag compartment cover-driver side
- 180 Air bag-passenger side
- 185 Air bag compartment cover-passenger side
- 190 Other air bag (specify)
- 195 Other air bag compartment cover (specify)
- 201 Front header
- 202 Rear header
- 203 Roof left side rail
- 204 Roof right side rail
- 205 Roof or convertible top
- 206 Roof maplight/console
- 207 Sunroof/components
- 208 Roll-bar
- 251 Floor (including toe pan)
- 252 Floor or console mounted transmission lever, incl console
- 253 Parking brake handle
- 254 Foot controls including parking brake
- 301 Backlight (rear window)
- 302 Backlight storage rack, door, etc.
- 303 Other rear object (specify)
- 401 Hand controls for braking/acceleration
- 402 Steering control devices (attached to OEM steering wheel)
- 403 Steering knob attached to steering wheel
- 405 Replacement steering wheel (i.e. reduced diameter)
- 406 Joy stick steering controls
- 407 Wheelchair tie-downs
- 408 Modification to seat belts, (specify)
- 409 Additional or relocated switches, (specify)
- 410 Raised roof
- 411 Wall mounted head rest (used behind wheel chair)
- 412 Other adaptive device (specify)
- 570 Same occupant contact (specify) [ex- knee]
- 571 Interior loose object (specify)
- 572 Seat LATCH points for child restraints
- 573 Grab handles
- 574 Engine shroud/cover
- 575 Seatback trays
- 576 Left forward upper quadrant

CONT1 position #16 (first contact point)

(continued)

CONT2 position #17 (second contact point)

- 577 Left forward lower quadrant
- 578 Left rear upper quadrant
- 579 Left rear lower quadrant
- 580 Left armrest/hardware in forward upper quadrant
- 581 Left armrest/hardware in forward lower quadrant
- 582 Left armrest/hardware in rear upper quadrant
- 583 Left armrest/hardware in rear lower quadrant
- 584 Right forward upper quadrant
- 585 Right forward lower quadrant
- 586 Right rear upper quadrant
- 587 Right rear lower quadrant
- 588 Right armrest/hardware in forward upper quadrant
- 589 Right armrest/hardware in forward lower quadrant
- 590 Right armrest/hardware in rear upper quadrant
- 591 Right armrest/hardware in rear lower quadrant
- 592 Child safety seat shell
- 593 Child safety seat harness
- 594 Unknown child safety seat component

Comment:

The indicated contact point is a component on the interior of the subject's vehicle for which there is physical evidence of contact by the CIREN subject. This is recorded on the NASS Interior Vehicle form, which is completed by the investigator during the inspection of the vehicle's interior. Some injuries have no contact points linked because there was no physical evidence of contact discovered during the vehicle inspection. Other injuries may have as many as two contact points associated with a single injury.

INTR1_LOC position #21 (first intrusion location)

INTR2_LOC position #22 (second intrusion location)

Short definition: The location within the subject's vehicle where an intrusion occurred that is thought to be associated with the current injury.

Code elements: two digits

Range: 11, 12, 13, 21, 22, 23, 31, 32, 33, 41, 42, 43, 51, 52, 53, null

Source: investigator determined

Code attributes:

Front Row

11 – Left

12 – Middle

13 – Right

Second Row

21 – Left

22 – Middle

23 – Right

(etc, Third, Fourth, and Fifth rows)

null - no intrusion linked

Comments:

The intrusion data are an element of the NASS Interior Vehicle form, completed by the crash investigator during the inspection of the vehicle's interior. When CIREN reconstructionists determine that an intrusion is related to an injury, the CIREN protocols enable linking the intrusion to the injury data. Some injuries have as many as ten intrusions linked; the two intrusions with the greatest magnitude are presented in this data set.

Intrusion location is presented in terms of seat rows, with each row divided into thirds. Users should note that the code scheme for intrusion location does not represent designated seating positions -- see the NASS-CDS Coding Manual for further information about the location of intrusions.

INTR1_COMP position #19 (first intruding component)

INTR2_COMP position #20 (second intruding component)

Short definition: The vehicle component or other object that intruded into the subject's vehicle and which is thought to be associated with the current injury.

Code elements: one or two digits

Range: n/a

Source: investigator determined

Code attributes:

Interior Components

- 1 - Steering assembly
- 2 - Instrument panel left
- 3 - Instrument panel center
- 4 - Instrument panel right
- 5 - Toe pan
- 6 - A (A1/A2)-pillar
- 7 - B-pillar
- 8 - C-pillar
- 9 - D-pillar
- 10 - Side panel - forward of the A1/A2-pillar
- 12 - Side panel - rear of the B-pillar
- 13 - Roof (or convertible top)
- 14 - Roof side rail
- 15 - Windshield
- 16 - Windshield header
- 17 - Window frame
- 18 - Floor pan (includes sill)
- 19 - Backlight header
- 20 - Front seat back
- 21 - Second seat back
- 22 - Third seat back
- 23 - Fourth seat back
- 24 - Fifth seat back
- 25 - Seat cushion
- 26 - Back door/panel (*e.g.*, tailgate)
- 27 - Other interior component (specify):
- 35 - Door/Forward upper quadrant
- 36 - Door/Forward lower quadrant

INTR1_COMP position #19 (first intruding component)

(continued)

INTR2_COMP position #20 (second intruding component)

Interior Components (continued)

37 - Door/Rear upper quadrant

38 - Door/Rear lower quadrant

41 - Door-Undetermined Location

Exterior Components

30 - Hood

31 - Outside surface of this vehicle (specify):

32 - Other exterior object in the environment (specify):

33 - Unknown exterior object

34 - Grab Handles

97 - Catastrophic

Other Component

96 - Multiple/Other severe intrusions

98 - Intrusion of exterior unlisted component(s)

null - no intrusion linked

Comments:

(See the introductory comments concerning intrusion under INTR1_LOC, above.)

INTR1_DIR position #25 (first intrusion direction)

INTR2_DIR position #26 (second intrusion direction)

Short definition: The direction that the intruding component moved as it intruded into the subject's vehicle.

Code elements: one digit

Range: null, 1 - 3, 7 - 9

Source: investigator determined

Code attributes:

- 1 - Vertical
- 2 - Longitudinal
- 3 - Lateral
- 7 - Catastrophic
- 8 - Multiple/Other Severe Intrusions
- 9 - Unknown

null - no intrusion linked

Comments:

(See the introductory comments concerning intrusion under INTR1_LOC, above.)

Intrusion measurements for a single component may be recorded for up to three directions (longitudinal, lateral and vertical) if the investigator believes the additional data are relevant.

INTR1_MAG position #23 (first intrusion magnitude)

INTR2_MAG position #24 (second intrusion magnitude)

Short definition: The magnitude of the intruding component's movement.

Code elements: one digit

Range: 0 - 9

Source: investigator determined

Code attributes:

0 - ≤ 2 cm

1 - ≥ 3 cm but < 8 cm

2 - ≥ 8 cm but < 15 cm

3 - ≥ 15 cm but < 30 cm

4 - ≥ 30 cm but < 46 cm

5 - ≥ 46 cm but < 61 cm

6 - ≥ 61 cm

7 - Catastrophic

8 - Multiple/Other Severe Intrusions

9 - Unknown

null - no intrusion linked

Comments:

See the introductory comments concerning intrusion under INTR1_LOC, above.

6.3 SCENARIO file

<u>Variable</u>	<u>SAS File Position</u>	<u>Short Description</u>
Identifiers =====		
CIRENID	1	CIREN case unique identifier (individual subject person)
CASEYEAR	8	calendar year when crash occurred
INJURYID	2	unique identifier for an individual injury
AISVERSION	56	version of AIS used to encode the injury
AISCODE	3	AIS code
ICS_NUMBER	9	Injury Causation Scenario (ICS) number
BT_LITE	10	BioTab Lite
BT_BR_INJ	11	BioTab body region injured
Caused by Other Injury =====		
C_INJURYID	4	causal injury identifier
C_AISCODE	5	causal injury AIS code
Injury Causation Scenario =====		
SOE	12	source of energy
S_ACCSEQ	14	NASSMAIN event sequence number for causal event
ICS_CONFID	13	ICS confidence
ICS_FACTOR1	15	first contributing factor
ICS_FACTOR2	16	second contributing factor
ICS_FACTOR3	17	third contributing factor
COMOR_FCT1	6	first comorbidity factor
COMOR_FCT2	7	second comorbidity factor
P_REG_MECH	18	primary regional mechanism
S_REG_MECH	19	secondary regional mechanism
Primary Involved Physical Component (IPC) =====		
PRIM_AREA	20	general area for IPC, primary
PRIM_IPC	21	IPC, primary
PRIM_ABLOC	22	location of air bag, primary
PRIM_BRC	23	body region contacted, primary
PRIM_CONF	24	IPC confidence, primary
PRIM_CI	25	critical intrusion, primary

Alternate IPC

=====

ALT_AREA	26	general area for IPC, alternate
ALT_IPC	27	IPC, alternate
ALT_ABLOC	28	location of air bag, alternate
ALT_BRC	29	body region contacted, alternate
ALT_CONF	30	IPC confidence, alternate
ALT_CI	31	critical intrusion, alternate

First Critical Primary IPC

=====

C1P_AREA	32	general area for IPC, first critical primary
C1P_IPC	33	IPC, first critical primary
C1P_ABLOC	34	location of air bag, first critical primary
C1P_BRC	35	body region contacted, first critical primary
C1P_CONF	36	IPC confidence, first critical primary
C1P_CI	37	critical intrusion, first critical primary

First Critical Alternate IPC

=====

C1A_AREA	38	general area for IPC, first critical alternate
C1A_IPC	39	IPC, first critical alternate
C1A_ABLOC	40	location of air bag, first critical alternate
C1A_BRC	41	body region contacted, first critical alternate
C1A_CONF	42	IPC confidence, first critical alternate
C1A_CI	43	critical intrusion, first critical alternate

Second Critical Primary IPC

=====

C2P_AREA	44	general area for IPC, second critical primary
C2P_IPC	45	IPC, second critical primary
C2P_ABLOC	46	location of air bag, second critical primary
C2P_BRC	47	body region contacted, second critical primary
C2P_CONF	48	IPC confidence, second critical primary
C2P_CI	49	critical intrusion, second critical primary

Second Critical Alternate IPC

=====

C2A_AREA	50	general area for IPC, second critical alternate
C2A_IPC	51	IPC, second critical alternate
C2A_ABLOC	52	location of air bag, second critical alternate
C2A_BRC	53	body region contacted, second critical alternate
C2A_CONF	54	IPC confidence, second critical alternate
C2A_CI	55	critical intrusion, second critical alternate

CIRENID position #1

Short definition: Unique identifier for a CIREN case (individual subject person).

Code elements: six or nine digits

Source: assigned by system

Range: N/A

Code attributes: N/A

Comments:

(none)

CASEYEAR position #8

Short definition: The calendar year in which the crash occurred.

Code elements: four digits

Source: official documents

Range: 2005 - 2014

Code Attributes: N/A

Comments:

(none)

INJURYID position #2

Short definition: Unique identifier for each specific injury.

Code elements: six or nine digits

Source: assigned by system

Range: N/A

Code attributes: N/A

Comment:

(none)

AISVERSION position #56

Short definition: Version of the AIS protocol used to encode the current injury.

Code elements: four digits

Source: assigned by system

Range: 1998, 2005

Code attributes: 1998, 2005

Comments:

The CIREN program adopted AIS-2005 as of 06/01/2010.

AISCODE position #3

Short definition: Abbreviated Injury Scale code for each specific injury.

Code elements: seven digits

Source: investigator determined

Range: structured code scheme always including seven digits

Code attributes: see the relevant AAAM/NHTSA Injury Coding Manuals

Comments:

The Abbreviated Injury Scale is an injury coding system developed by the Association for the Advancement of Automotive Medicine (AAAM)^{3,4}. CIREN used AIS version 1990 Update 1998 for all crashes occurring prior to June 1, 2010. Crash occupants captured in CIREN on June 1, 2010 and later have been coded using the AIS 2005 Update 2008 version.

ICS_NUMBER position #9

Short definition: There may be either one or two Injury Causation Scenarios for each specific injury. ICS_NUMBER indicates whether this record is the first or second scenario.

Code elements: one digit

Source: assigned by system

Range: -9, 1, 2

Code attributes:

- 9 - Pre BioTabLite, BioTab not done
- 1 - first Injury Causation Scenario
- 2 - second Injury Causation Scenario

Comment:

There will be either one or two scenario records for each specific injury. For very early cases, the BioTabLite coding protocol was not available and minor injuries were coded according to the abbreviated version of the NASS injury coding protocol – the limited data from this “short form” coding have been translated to the BioTab format, with most BioTab items coded “-9”. For ICS details please reference Schneider et al. 2011.

BT_LITE position #10

Short definition: Minor injuries are entered under an abbreviated protocol called BioTabLite; this variable flags injuries that are coded per the BioTabLite protocol.

Code elements: one digit

Source: assigned by system

Range: -9, 0, 1

Code attributes:

- 9 - Pre BioTabLite, BioTab not done
- 0 - no, not BioTabLite (full BioTab protocol observed)
- 1 - yes, BioTabLite

Comment:

The BioTabLite protocol was introduced in phases beginning in mid-2007, with full availability for cases with crash date 01/01/2008 or later. For cases with an earlier crash date, data for injuries that would later be covered by BioTabLite were entered under an abbreviated version of the NASS-CDS injury coding protocol (the “short form”).

BT_BR_INJ position #11

Short definition: BioTab Body Region Injured; the human body is divided into various regions for the purposes of developing an Injury Causation Scenario. BT_BR_INJ contains a code to indicate the Biomechanical Body Region in which the current injury is located.

Code elements: one, two or four digits

Source: investigator determined

Range: -9999, -9, 1 - 20

Code attributes:

-9999 - ICS Unknown

-9 - Pre BioTabLite, BioTab not done

1 - Head/Face

2 - Neck

3 - Cervical Spine

4 - Shoulder

5 - Arm

6 - Elbow

7 - Forearm

8 - Wrist

9 - Hand

10 - Thorax

11 - Thoracic Spine

12 - Abdomen

13 - Lumbar Spine

14 - Pelvis

15 - Hip

16 - Thigh

17 - Knee

18 - Leg

19 - Ankle

20 - Foot

Comment:

The BioTab body regions were defined by the CIREN project for the purpose of developing an Injury Causation Scenario. All of the various anatomical structures in the human body are assigned to a BioTab body region. The BioTab body regions were developed from the AIS body region assignments².

C_INJURYID position #4

Short definition: A given specific injury may be judged to have been caused by another specific injury; C_INJURYID contains the system-assigned INJURYID for the other injury that caused the current injury.

Code elements: one, five, six or nine digits

Source: assigned by system

Range: 0, xxxxx, xxxxxx, xxxxxxxxx

Code attributes:

0 - no causal injury (i.e., the current injury was not caused by another injury)

xxxxx,

xxxxxx,

xxxxxxxxx - INJURYID for the causal injury

Comment:

When it is determined that the current specific injury was caused by another specific injury, the investigator designates the causal injury and that injury's system-assigned INJURYID is linked to the Injury Causation Scenario for the current injury.

C_AISCODE position #5

Short definition: A given specific injury may be judged to have been caused by another specific injury; C_AISCODE contains the AIS code for the other injury that caused the current injury.

Code elements: one or seven digits

Source: investigator determined

Range: 0, xxxxxxx

Code attributes:

0 - no causal injury (i.e., the current injury was not caused by another injury)
xxxxxxx - AIS code for the causal injury (see C_INJURYID)

Comment:

When it is determined that the current specific injury was caused by another specific injury, the investigator designates the causal injury. This variable contains the AIS code for the causal injury.

SOE position #12

Short definition: The source of the energy that caused the current specific injury.

Code elements: one or four digits

Source: investigator determined

Range: -9999, -8887, -9, 1 - 4

Code attributes:

- 9999 - Injury Causation Scenario unknown
- 8887 - N/A (this injury was caused by another injury)
- 9 - Pre BioTabLite, BioTab not done
- 1 - crash
- 2 - air bag
- 3 - pretensioner
- 4 - fire

Comment:

(none)

S_ACCSEQ position #14

Short definition: If a crash event was the source of energy that caused an injury (see SOE), that event's chronological sequence number from among all events in the crash.

Code elements: one, two or four digits

Source: assigned by system

Range: -9999, -8888, -8887, -9, 0, 1 - 30

Code attributes:

- 9999 - Injury Causation Scenario unknown
- 8888 - not applicable; this injury not caused by the energy from a crash event
- 8887 - not applicable; this injury caused by another injury
- 9 - Pre BioTabLite, BioTab not done
- 0 - no event selected
- 1 - 30 - NASS event chronological sequence number

Comment:

See the NASS-CDS Coding and Editing Manual for information concerning the protocol that governs the coding and entry of crash events.

S_ACCSEQ enables linkage to the NASS event structure via ACCSEQ in the Event file.

ICS_CONFID position #13

Short definition: The team's confidence about the Injury Causation Scenario.

Code elements: one or 4 digits

Source: Team determined

Range: -9999, -9, 1 - 3

Code attributes:

- 9999 - Injury Causation Scenario unknown
- 9 - Pre BioTabLite, BioTab not done
- 1 - Certain
- 2 - Probable
- 3 - Possible

Comment:

(none)

ICS_FACTOR1 position #15 (Injury Causation Scenario Contributing Factor, entry #1)
 ICS_FACTOR2 position #16 (Injury Causation Scenario Contributing Factor, entry #2)
 ICS_FACTOR3 position #17 (Injury Causation Scenario Contributing Factor, entry #3)

Short definition: Factors that contributed to the existence/severity of this injury that were identified during the development of the Injury Causation Scenario.

Code elements: null, one, two or four digits

Source: Team determined

Range: -9999, -8887, -9, 0 - 21, null

Code attributes:

- 9999 - ICS Unknown
- 8887 - Caused by other injury, ICS Contributing Factors not assessed
- 9 - Pre BioTabLite, BioTab not done
- 0 - BioTabLite injury, ICS Contributing Factors not assessed
- 1 - Elderly - General *[retired 11-01-2009]*
- 2 - Occupant position/posture (specify) *[retired 11-01-2009]*
- 3 - Improper restraint use (specify) *[retired 11-01-2009]*
- 4 - Tall stature *[retired 11-01-2009]*
- 5 - Short stature *[retired 11-01-2009]*
- 7 - Intrusion
- 8 - Partial ejection
- 9 - CoMorbidity (pick)
- 10 - Child Restraint System improperly used (Specify)
- 11 - Unbelted Case Occupant
- 12 - Pretensioner
- 13 - Other (specify)
- 14 - High Delta V
- 15 - None
- 16 - Loose Cargo
- 17 - Possible Late Air Bag Deployment (specify)
- 18 - Seat belt payout due to load limiter
- 19 - Unbelted Other Occupant
- 20 - Elderly - (specify)
- 21 - Seat Belt Interaction (specify)
- null - no additional ICS factor (ICS_FACTOR2 and ICS_FACTOR3 only)

Comment:

Some scenarios include as many as five entries for Contributing Factors; the first three are presented in this data file. Contributing Factors are coded when the coded factor has increased the likelihood of the injury and/or increased the severity of the injury.

COMOR_FCT1 position #6 (First comorbidity cited as a contributing factor)

COMOR_FCT2 position #7 (Second comorbidity cited as a contributing factor)

Short definition: One aspect of Contributing Factors involves designating a comorbidity as a contributing factor; the code for that comorbidity, if designated, is presented here.

Code elements: one, three or four digits

Source: investigator determined

Range: -9999, -8887, -9, -7, 0, 101 - 1601

Code attributes:

-9999 - ICS unknown

-8887 - Caused by other injury, Contributing Factors not assessed

-9 - Pre BioTabLite, BioTab not done

-7 - Contributing Factors assessed, no Comorbidity indicated (COMOR_FCT1);

-7 - No second Comorbidity (COMOR_FCT2)

0 - BioTabLite injury, Contributing Factors not assessed

101 - GENERAL * RECENT DELIVERY

102 - GENERAL * PREGNANCY

201 - OTHER * UNKNOWN

202 - OTHER * NONE

203 - OTHER * OTHER

301 - CARDIAC DISEASE * HISTORY OF CARDIAC SURGERY

302 - CARDIAC DISEASE * CORONARY ARTERY DISEASE

303 - CARDIAC DISEASE * CONGESTIVE HEART FAILURE

304 - CARDIAC DISEASE * COR PULMONALE

305 - CARDIAC DISEASE * MYOCARDIAL INFARCTION

306 - CARDIAC DISEASE * HYPERTENSION

401 - DIABETES * INSULIN DEPENDENT

402 - DIABETES * NON-INSULIN DEPENDENT (ON ORAL MEDICATION)

501 - GASTROINTESTINAL DISEASE * PEPTIC ULCER DISEASE

502 - GASTROINTESTINAL DISEASE * GASTRIC OR ESOPHAGEAL VARICES

503 - GASTROINTESTINAL DISEASE * PANCREATITIS

504 - GASTROINTESTINAL DISEASE * INFLAMMATORY BOWEL DISEASE

601 - HEMATOLOGIC DISORDERS * ACQUIRED COAGULOPATHY

602 - HEMATOLOGIC DISORDERS * BLOOD THINNER THERAPY

603 - HEMATOLOGIC DISORDERS * HEMOPHILIA

604 - HEMATOLOGIC DISORDERS * PRE-EXISTING ANEMIA

701 - PSYCHIATRIC DISORDERS * HISTORY OF PSYCHIATRIC DISORDERS

801 - IMMUNOSUPPRESSED THERAPY * TRANSPLANTS

802 - IMMUNOSUPPRESSED THERAPY * HIV/AIDS

803 - IMMUNOSUPPRESSED THERAPY * ROUTINE STEROID USE

804 - IMMUNOSUPPRESSED THERAPY * ACTIVE CHEMOTHERAPY

901 - LIVER DISEASE * BILIRUBIN >2MG O/O (ON ADMISSION)

902 - LIVER DISEASE * DOCUMENTED HISTORY OF CIRRHOSIS

1001 - MALIGNANCY * UNDERGOING CURRENT THERAPY FOR MALIGNANCY

COMOR_FCT1 position #6
COMOR_FCT2 position #7

(continued)

1002 - MALIGNANCY * CONCURRENT OR EXISTENCE OF METASTASIS
1101 - MUSCULOSKELETAL * OSTEOPOROSIS
1102 - MUSCULOSKELETAL * RHEUMATOID ARTHRITIS
1103 - MUSCULOSKELETAL * SYSTEMIC LUPUS ERYTHEMATOUS
1104 - MUSCULOSKELETAL * PRIOR JOINT REPLACEMENT
1105 - MUSCULOSKELETAL * PRIOR SURGICAL IMPLANTS
1201 - NEUROLOGIC * SPINAL CORD INJURY
1202 - NEUROLOGIC * MULTIPLE SCLEROSIS
1203 - NEUROLOGIC * ALZHEIMERS DISEASE
1204 - NEUROLOGIC * SEIZURES
1205 - NEUROLOGIC * CHRONIC DEMYELINATING DISEASE
1206 - NEUROLOGIC * CHRONIC DEMENTIA
1207 - NEUROLOGIC * ORGANIC BRAIN SYNDROME
1208 - NEUROLOGIC * PARKINSONS DISEASE
1209 - NEUROLOGIC * CVA/HEMIPARESIS (STROKE WITH RESIDUAL)
1301 - PULMONARY * DOCUMENTED PRIOR HX OF PULM DX W/ ONGOING ACTIVE TX
1302 - PULMONARY * ASTHMA
1303 - PULMONARY * CHRONIC OBSTRUCTIVE PULMONARY DISEASE
1304 - PULMONARY * CHRONIC PULMONARY CONDITION
1401 - RENAL DISEASE * SERUM CREATININE >2MG O/O (ON ADMISSION)
1402 - RENAL DISEASE * DIALYSIS (EXCLUDES TRANSPLANT PATIENTS)
1501 - SUBSTANCE ABUSE * CHRONIC DRUG USER
1502 - SUBSTANCE ABUSE * CHRONIC ALCOHOL USER
1601 - OBESITY * OBESITY

Comment:

(none)

P_REG_MECH position #18 (Primary Regional Mechanism)

S_REG_MECH position #19 (Secondary Regional Mechanism)

Short definition: The physical mechanism that caused the injury; “Regional Mechanism” refers to the BioTab Body Region (see BT_BR_INJ).

Code elements: one, two or four digits

Source: investigator determined

Range: -9999, -8887, -9, 0, 1 - 14, 16 - 37

Code attributes:

-9999 - ICS Unknown (P_REG_MECH)	19 - Torsion
-9999 - mechanism unknown (S_REG_MECH)	20 - Rate of Compression
-8887 - Cause by other injury	23 - Dorsiflexion
-9 - Pre BioTabLite, BioTab not done	24 - Plantarflexion
0 - BioTabLite, Mechanism not assessed	26 - Buckling
1 - Compression	27 - Bending
2 - Shear	28 - Compression and rate of compression
3 - Puncture	29 - Medial/Lateral Bending
4 - Cutting	30 - Dorsiflexion/Plantarflexion
5 - Crushing	31 - Inversion/Eversion
6 - Thermal	33 - Compression/Flexion
7 - Chemical	34 - Compression/Extension
8 - Linear Acceleration	35 - Compression/Lateral Bending
9 - Angular Acceleration	36 - Extension
10 - Tension	37 - Flexion
13 - Flexion/Extension	38 - Acceleration
14 - Lateral Bending	
16 - Abduction/Adduction	
17 - Over Pressure	
18 - Buckling/Bending	

Comment:

(none)

PRIM_AREA position #20 (Area for Primary IPC)

Short Definition: The general area where the IPC is located.

Code Elements: one, two or four digits

Source: Investigator determined

Range: -9999, -8887, 0, 1 - 16, 20

Code Attributes:

- 9999 - Unknown
- 8887 - Caused by other injury
- 0 - No entry (see discussion of Primary/Alternate and First Critical/Second Critical)
- 1 - Front
- 2 - Left Side
- 3 - Right Side
- 4 - Interior
- 5 - Air Bag
- 6 - Roof
- 7 - Floor
- 8 - Rear
- 9 - Adaptive (Assistive) Driving Equipment
- 10 - Exterior of Occupant's Motor Vehicle
- 11 - Exterior of Other Motor Vehicle
- 12 - Other Object in the Environment
- 13 - Noncontact Injury
- 14 - Injured, Unknown Source
- 15 - Left Door Panel
- 16 - Right Door Panel
- 20 - Other Injury

Comment:

Physical Components are grouped according to the general area where they are located.

The "injury source" data from the pre-BioTabLite era have been translated into the primary IPC fields for this data set.

The code "0 - no entry" is used to accommodate the multiplicity of options for designating the interaction between a person and the Involved Physical Component (IPC).

PRIM_IPC position #21 (Primary IPC)

Short Definition: The Involved Physical Component (IPC) that was contacted by the subject and which was the point through which the load was applied to the Body Region Contacted (BRC).

Code Elements: one to four digits

Source: Investigator Determined

Range: -9999, -8887, 0 - 699

Code Attributes:

-9999 ICS unknown
 -8887 Caused by other injury
 0 No entry (see discussion of Primary/Alternate and First Critical/Second Critical)

FRONT

- 1 Windshield
- 2 Mirror
- 3 Sunvisor
- 4 Steering wheel rim
- 5 Steering wheel hub/spoke
- 6 Steering wheel (combination of codes 004 and 005)
- 7 Steering column, transmission selector lever, other attachment
- 8 Cellular telephone or CB radio
- 9 Add on equipment (e.g., tape deck, air conditioner)
- 10 Left instrument panel and below
- 11 Center instrument panel and below
- 12 Right instrument panel and below
- 13 Glove compartment door
- 14 Knee bolster
- 15 Win. incl. 1/+:fr header, A(A1/A2)-pillar,instr. panel,mirror,or steering assembly(driver)
- 16 Win. incl. 1/+:fr header, A(A1/A2)-pillar,instr. panel,or mirror(passenger)
- 17 Windshield reinforced by exterior object(specify)
- 19 Other front object (specify)
- 20 Sunvisor reinforced by front header
- 21 Left Instrument Panel
- 22 Center Instrument Panel
- 23 Right Instrument Panel
- 24 Left lower instrument panel (includes knee bolster)
- 25 Center lower instrument panel (includes knee bolster)
- 26 Right lower instrument panel (includes knee bolster)

LEFT SIDE

- 51 Left side interior surface, excluding hardware or armrests
- 52 Left side hardware or armrest
- 53 Left A(A1/A2)-pillar
- 54 Left B-pillar
- 55 Other left pillar(specify)

PRIM_IPC position #21 (Primary IPC)

(continued)

- 56 Left side window glass
- 57 Left side window frame
- 58 Left side window sill
- 59 Lt side win. glass incl. 1/+:frame,win. sill,A(A1/A2)-pillar,B-pillar,or roof side rail
- 60 Other left side object (specify)
- 61 Left side glass (laminated) re-inforced by exterior object (specify)
- 62 Left side panel forward of A1/A2 pillar
- 63 Left side panel rear of B pillar

LEFT DOOR PANEL

- 73 Left forward upper quadrant
- 74 Left forward lower quadrant
- 75 Left rear upper quadrant
- 76 Left rear lower quadrant
- 77 Left armrest/hardware in forward upper quadrant
- 78 Left armrest/hardware in forward lower quadrant
- 79 Left armrest/hardware in rear upper quadrant
- 80 Left armrest/hardware in rear lower quadrant

RIGHT SIDE

- 101 Right side interior surface, excluding hardware or armrests
- 102 Right side hardware or armrest
- 103 Right A (A1/A2)-pillar
- 104 Right B-pillar
- 105 Other right pillar (specify)
- 106 Right side window glass
- 107 Right side window frame
- 108 Right side window sill
- 109 Rt side win. glass incl. 1/+:frame, win. sill, A(A1/A2)-pillar, B-pillar, or roof side rail
- 110 Other right side object (specify)
- 111 Right side glass (laminated) reinforced by exterior object (specify)
- 112 Right side panel forward of A1/A2 pillar
- 113 Right side panel rear of B pillar

RIGHT DOOR PANEL

- 121 Right forward upper quadrant
- 122 Right forward lower quadrant
- 123 Right rear upper quadrant
- 124 Right rear lower quadrant
- 125 Right armrest/hardware in forward upper quadrant
- 126 Right armrest/hardware in forward lower quadrant
- 127 Right armrest/hardware in rear upper quadrant
- 128 Right armrest/hardware in rear lower quadrant

INTERIOR

- 151 Seat, back support
- 152 Belt restraint webbing/buckle
- 153 Belt restraint B-pillar or door frame attachment point
- 154 Other restraint system component (specify)
- 155 Head restraint system

PRIM_IPC position #21 (Primary IPC)

(continued)

- 160 Other occupants (specify)
- 161 Interior loose objects (specify)
- 162 Child safety seat (specify)
- 163 Other interior object (specify)
- 164 Center console first row
- 165 Center console second row
- 166 Center console other row
- 167 Fold down armrest first row
- 168 Fold down armrest second row
- 169 Fold down armrest other row
- 271 Child safety seat shell
- 272 Child safety seat harness system
- 273 Unknown child safety seat component
- 570 Same occupant contact (specify) [ex- knee]
- 571 Interior loose object (specify)
- 572 Seat LATCH points for child restraints
- 573 Grab handles
- 574 Engine shroud/cover
- 575 Seatback trays
- 576 Cargo in vehicle

ROOF

- 201 Front header
- 202 Rear header
- 203 Roof left side rail
- 204 Roof right side rail
- 205 Roof or convertible top
- 206 Roof maplight/console
- 207 Sunroof/components
- 208 Roll-bar

FLOOR

- 251 Floor (including toe pan)
- 252 Floor or console mounted transmission lever, including console
- 253 Parking brake handle
- 254 Foot controls including parking brake

REAR

- 301 Backlight (rear window)
- 302 Backlight storage rack, door, etc
- 303 Other rear object (specify)

AIR BAG

- 320 Air bag
- 321 Air bag and eyewear
- 322 Air bag and jewelry
- 323 Air bag and object held
- 324 Air bag and object in mouth
- 325 Air bag compartment cover
- 326 Air bag compartment cover and eyewear

PRIM_IPC position #21 (Primary IPC)

(continued)

- 327 Air bag compartment cover and jewelry
- 328 Air bag compartment cover and object held
- 329 Air bag compartment cover and object in mouth

ADAPTIVE (ASSISTIVE) DRIVING EQUIPMENT

- 401 Hand controls for braking/acceleration
- 402 Steering control devices (attached to OEM steering wheel)
- 403 Steering knob attached to steering wheel
- 405 Replacement steering wheel (i.e., reduced diameter)
- 406 Joy stick steering controls
- 407 Wheelchair tie-downs
- 408 Modification to seat belts (specify)
- 409 Additional or relocated switches (specify)
- 410 Raised roof
- 411 Wall mounted head rest (used behind wheel chair)
- 412 Other adaptive device (specify)

EXTERIOR OF OCCUPANT'S MOTOR VEHICLE

- 451 Hood
- 452 Outside hardware (e.g., outside mirror, antenna)
- 453 Other exterior surface or tires (specify)
- 454 Unknown exterior objects, this occupant's vehicle

EXTERIOR OF OTHER MOTOR VEHICLE

- 501 Front bumper
- 502 Hood edge
- 503 Other front of other vehicle (specify)
- 504 Hood
- 505 Hood ornament
- 506 Windshield, roof rail, A-pillar
- 507 Side surface
- 508 Side mirrors
- 509 Other side protrusions (specify)
- 510 Rear surface
- 511 Undercarriage
- 512 Tires and wheels
- 513 Other exterior of other motor vehicle (specify)
- 514 Unknown exterior of other motor vehicle

OTHER OBJECT IN THE ENVIRONMENT

- 551 Ground
- 552 Tree
- 553 Pole
- 554 Traffic barrier (includes: jersey barrier, guardrails, etc.)
- 598 Other object in the environment (specify)
- 599 Unknown object in the environment (specify)

PRIM_IPC position #21 (Primary IPC)

(continued)

NONCONTACT INJURY

- 601 Fire in vehicle
- 602 Flying glass
- 603 Other noncontact injury source (specify)
- 604 Air bag exhaust gases

- 697 Injured, unknown source
- 699 Other Injury

Comment:

The “injury source” data from the pre-BioTabLite era have been translated into the primary IPC fields for this data set.

The code “0 - no entry” is used to accommodate the multiplicity of options for designating the interaction between a person and the IPC. For IPC details please see Schneider et al. 2011.

PRIM_ABLOC position #22

Short Definition: When the Primary IPC is an air bag or air bag-related component (IPC codes 320 - 329), the location of that air bag.

Code Elements: one, two or four digits

Source: Investigator determined

Range: null, -8887, 1 - 5, 8 - 11

Code Attributes:

- null - the IPC is not an air bag or air bag module component

- 8887 - caused by other injury

- 1 - steering wheel hub

- 2 - top instrument panel

- 3 - mid instrument panel

- 4 - bottom instrument panel

- 5 - seat back

- 8 - door/panel

- 9 - roof side rail

- 10 - other

- 11 - unknown

Comment:

Users should note that the exact location of the air bag is not designated. For most injuries, the indicated air bag is associated with the subject's seat position, but this is not necessarily true. There are instances where, e.g., an unbelted driver (seated on the left) moves across the front seat row and contacts the roof rail air bag on the right; left vs. right roof rail cannot be determined from this variable. Analysts should not make any assumptions in this connection.

PRIM_BRC position #23

Short Definition: The Body Region Contacted (BRC) that interacted with the Primary IPC.

Code Elements: one, two or four digits

Source: Investigator determined

Range: -9999, -8887, -9, 0, 1 - 20

Code Attributes:

-9999 - ICS Unknown	10 - Thorax
-8887 - N/A, caused by other injury	11 - Thoracic Spine
-9 - Pre BioTabLite, BioTab not done	12 - Abdomen
0 - no entry	13 - Lumbar Spine
1 - Head/Face	14 - Pelvis
2 - Neck	15 - Hip
3 - Cervical Spine	16 - Thigh
4 - Shoulder	17 - Knee
5 - Arm	18 - Leg
6 - Elbow	19 - Ankle
7 - Forearm	20 - Foot
8 - Wrist	
9 - Hand	

Comment:

The code “0 - no entry” is used to accommodate the multiplicity of options for designating the interaction between a person and the Involved Physical Component (IPC).

PRIM_CONF position #24

Short Definition: The team's degree of confidence in the Primary IPC.

Code Elements: one or four digits

Source: Team Determined

Range: -9999, -8887, 0, 1 - 3

Code Attributes:

-9999 - Injury Causation Scenario unknown

-8887 - caused by other injury

0 - no entry

1 - Certain

2 - Probable

3 - Possible

Comment:

For pre-BioTabLite injuries, the Level of Confidence indicated for the old-style "injury source" coding has been translated to this field.

The code "0 - no entry" is used to accommodate the multiplicity of options for designating the interaction between a person and the Involved Physical Component (IPC).

PRIM_CI position #25

Short Definition: Critical Intrusion associated with the Primary IPC.

Code Elements: one, two or four digits

Source: Team determined

Range: -8887, 0 , 1 - 27, 30 - 39, 41, 96, 98, 99

Code Attributes:

-8887 Caused by other injury
0 No critical intrusion
01 Steering Assembly
02 Instrument panel left
03 Instrument panel center
04 Instrument panel right
05 Toe pan
06 A (A1/A2)-pillar
07 B-pillar
08 C-pillar
09 D-pillar
10 Side panel - forward of the A1/A2 pillar
11 Door panel (side)
12 Side panel - rear of the B-pillar
13 Roof (or convertible top)
14 Roof side rail
15 Windshield
16 Windshield header
17 Window frame
18 Floor pan (includes sill)
19 Backlight header
20 Front seat back
21 Second seat back
22 Third seat back
23 Fourth seat back
24 Fifth seat back
25 Seat cushion
26 Back door/panel (e.g., tailgate)
27 Other interior component (specify):
30 Hood
31 Outside surface of this vehicle (specify):

PRIM_CI (Critical Intrusion for the Primary IPC)

(continued)

- 32 Other exterior object in the environment (specify):
- 33 Unknown exterior object
- 35 Door/Forward upper quadrant
- 36 Door/Forward lower quadrant
- 37 Door/Rear upper quadrant
- 38 Door/Rear lower quadrant
- 39 Multiple other severe intrusions
- 41 Door- Undetermined Location
- 96 Multiple/Other severe intrusions
- 98 Intrusion of unlisted component(s) (specify):
- 99 Unknown

Comment:

A Critical Intrusion is an intrusion into the occupant compartment that was required in order for the BRC to make contact with the IPC. Critical Intrusion is not based solely on the magnitude of the intrusion, but is based on expert consensus as to whether the occupant's BRC could have made contact with the IPC if the intrusion had not occurred. The designation of a Critical Intrusion considers occupant size, restraint status, and trajectory. If the occupant would normally make contact with the intruded component in an undeformed vehicle with a similar crash pulse, the intrusion is not critical, and coding the intrusion as a Contributing Factor is warranted (see ICS_FACTOR1).

ALT_AREA position #26 (Area for Alternate IPC)
ALT_IPC position #27 (Alternate IPC)
ALT_ABLOC position #28 (Air Bag Location for Alternate IPC)
ALT_BRC position #29 (BRC for Alternate IPC)
ALT_CONF position #30 (Confidence for Alternate IPC)
ALT_CI position #31 (Critical Intrusion for Alternate IPC)

The cluster of variables with prefix “ALT” documents the Alternate IPC; see the discussion of the Primary IPC for attribute definitions.

C1P_AREA position #32 (Area for First Critical Primary IPC)

C1P_IPC position #33 (First Critical Primary IPC)

C1P_ABLOC position #34 (Air Bag Location for First Critical Primary IPC)

C1P_BRC position #35 (BRC for First Critical Primary IPC)

C1P_CONF position #36 (Confidence for First Critical Primary IPC)

C1P_CI position #37 (Critical Intrusion for First Critical Primary IPC)

The cluster of variables with prefix “C1P” documents the First Critical Primary IPC; see the discussion of the Primary IPC for attribute definitions.

C1A_AREA position #38 (Area for First Critical Alternate IPC)

C1A_IPC position #39 (First Critical Alternate IPC)

C1A_ABLOC position #40 (Air Bag Location for First Critical Alternate IPC)

C1A_BRC position #41 (BRC for First Critical Alternate IPC)

C1A_CONF position #42 (Confidence for First Critical Alternate IPC)

C1A_CI position #43 (Critical Intrusion for First Critical Alternate IPC)

The cluster of variables with prefix “C1A” documents the First Critical Alternate IPC; see the discussion of the Primary IPC for attribute definitions.

C2P_AREA position #44 (Area for Second Critical Primary IPC)
C2P_IPC position #45 (Second Critical Primary IPC)
C2P_ABLOC position #46 (Air Bag Location for Second Critical Primary IPC)
C2P_BRC position #47 (BRC for Second Critical Primary IPC)
C2P_CONF position #48 (Confidence for Second Critical Primary IPC)
C2P_CI position #49 (Critical Intrusion for Second Critical Primary IPC)

The cluster of variables with prefix “C2P” documents the Second Critical Primary IPC; see the discussion of the Primary IPC for attribute definitions.

6.4 Injuryanalysis file

SAS File		
<u>Variable</u>	<u>Position</u>	<u>Short Description</u>
Identifiers		
=====		
CIRENID	1	CIREN case unique identifier (individual subject person)
LINENO	2	sequential order number of text lines
CASEYEAR	4	calendar year when crash occurred
Injury Analysis		
=====		
INJTEXT	3	Injury Analysis narrative text

CIRENID position #1

Short definition: Unique identifier for a CIREN case (individual subject person). Code elements: six or nine digits

Code elements: six or nine digits

Source: assigned by system

Range: N/A

Code attributes: N/A

Comments:

(none)

LINENO position #2

Short definition: Sequential order number of text lines

Code elements: One or two digits

Source: assigned by system

Range: N/A

Code attributes: N/A

Comments:

Assembling the LINENO in ascending order will reconstruct the complete CIREN Injury Analysis.

CASEYEAR position #4

Short definition: The calendar year in which the crash occurred.

Code elements: four digits

Source: official documents

Range: 2005 - 2014

Code Attributes: N/A

Comments:

(none)

INJTEXT position #3

Short definition: Injury Analysis narrative text

Code elements: N/A

Source: Generated by CIREN

Range: N/A

Code attributes: N/A

Comments:

(none)

6.5 GV (general vehicle) file

Contents (SAS Proc Contents, alphabetical order)

#	Variable	Type	Len	Label
1	ACCSEQDV	Num	3	ACCIDENT SEQUENCE NO FOR HIGHEST DELTA V
2	ACCTYPE	Num	3	ACCIDENT TYPE
3	ALCTEST	Num	3	ALCOHOL TEST RESULT FOR DRIVER
4	ALIGNMNT	Num	3	ROADWAY ALIGNMENT
5	ANGOTHER	Num	3	HEADING ANGLE FOR OTHER VEHICLE
6	ANGTHIS	Num	3	HEADING ANGLE FOR THIS VEHICLE
7	AOPSVEH	Num	3	AOPS VEHICLE
8	BAGDEPFV	Num	3	AIR BAG DEPLOYMENT, FIRST SEAT FRONTAL
9	BAGDEPOV	Num	3	AIR BAG DEPLOYMENT, OTHER
10	BAREQSP	Num	3	BARRIER EQUIVALENT SPEED
11	BODYTYPE	Num	3	VEHICLE BODY TYPE
12	CARGOWGT	Num	3	VEHICLE CARGO WEIGHT
13	CASENO	Num	8	CASE SEQUENCE NUMBER
14	CASEYEAR	Num	3	CASEYEAR
15	CLIMATE	Num	3	WEATHER
16	CONDTREE	Num	3	POST COLLISION CONDITION OF TREE OR POLE
17	CURBWGT	Num	4	VEHICLE CURB WEIGHT
18	DOCTRAJ	Num	3	DOCUMENTATION OF TRAJECTORY DATA
19	DRINKING	Num	3	POLICE REPORTED ALCOHOL PRESENCE
20	DRIVDIST	Num	3	DRIVER'S DISTRACTION/INATTENTION TO DRIVING
21	DRPRES	Num	3	DRIVER PRESENCE IN VEHICLE
22	DRRACE	Num	3	DRIVER'S RACE/ETHNIC ORIGIN
23	DRUGS	Num	3	REPORTED OTHER DRUG
24	DRZIP	Num	4	DRIVER'S ZIP CODE
25	DVBASIS	Num	3	BASIS FOR TOTAL DELTA V (HIGHEST)
26	DVCONFID	Num	3	CONFIDENCE IN RECONSTRUCTION
27	DVEST	Num	3	ESTIMATED HIGHEST DELTA V
28	DVLAT	Num	3	LATERAL COMPONENT OF DELTA V
29	DVLONG	Num	3	LONGITUDINAL COMPONENT OF DELTA V
30	DVTOTAL	Num	3	TOTAL DELTA V
31	ENERGY	Num	4	ENERGY ABSORPTION
32	ETHNICIT	Num	3	ETHNICITY
33	FOVERRIDE	Num	3	FRONT OVERRIDE/UNDERRIDE THIS VEHICLE
34	IMPACTSP	Num	3	IMPACT SPEED
35	INSPTYPE	Num	3	TYPE OF VEHICLE INSPECTION

#	Variable	Type	Len	Label
36	INTEROLL	Num	3	INTERRUPTED ROLLOVER
37	LANES	Num	3	NUMBER OF LANES
38	LGTCND	Num	3	LIGHT CONDITIONS
39	MAKE	Num	3	VEHICLE MAKE
40	MANEUVER	Num	3	ATTEMPTED AVOIDANCE MANEUVER
41	MODEL	Num	3	VEHICLE MODEL
42	MODELYR	Num	4	VEHICLE MODEL YEAR
43	OCCFORMS	Num	3	NUMBER OF OCCUPANT FORMS SUBMITTED
44	OCUPANTS	Num	3	NUMBER OF OCCUPANTS THIS VEHICLE
45	PREEVENT	Num	3	INITIAL CRITICAL (PRECRASH) EVENT
46	PREILOC	Num	3	PRE-IMPACT LOCATION
47	PREISTAB	Num	3	PRE-IMPACT STABILITY
48	PREMOVE	Num	3	PRE-EVENT MOVEMENT PRIOR REC CRIT EVENT
49	PROFILE	Num	3	ROADWAY PROFILE
50	PROLLMAN	Num	3	PRE ROLLOVER MANEUVER
51	PSU	Num	3	PRIMARY SAMPLING UNIT NUMBER
52	RACE	Num	3	RACE
53	RELINTER	Num	3	RELATION TO JUNCTION
54	ROLINDIR	Num	3	DIRECTION OF INITIAL ROLL
55	ROLINLOC	Num	3	LOCATION OF ROLLOVER
56	ROLINTYP	Num	3	ROLLOVER INITIATION TYPE
57	ROLLDIST	Num	3	ESTIMATED DISTANCE OF ROLLOVER
58	ROLLOBJ	Num	3	ROLLOVER INITIATION OBJECT CONTACTED
59	ROLLOVER	Num	3	ROLLOVER
60	ROVERRIDE	Num	3	REAR OVERRIDE/UNDERRIDE THIS VEHICLE
61	SPECOTH	Num	3	OTHER DRUG: SPECIMEN TEST RESULTS
62	SPLIMIT	Num	3	SPEED LIMIT
63	SURCOND	Num	3	ROADWAY SURFACE CONDITION
64	SURTYPE	Num	3	ROADWAY SURFACE TYPE
65	TOWHITCH	Num	3	TOWED TRAILING UNIT
66	TOWPAR	Num	3	POLICE REPORTED VEHICLE DISPOSITION
67	TRAFCONT	Num	3	TRAFFIC CONTROL DEVICE
68	TRAFFLOW	Num	3	TRAFFICWAY FLOW
69	TRANSTAT	Num	3	TRANSPORT STATUS
70	TRAVELSP	Num	3	POLICE REPORTED TRAVEL SPEED
71	TRCTLFCT	Num	3	TRAFFIC CONTROL DEVICE FUNCTIONING
72	TRIPLOC	Num	3	LOC. ON VEH. WHERE INIT TRIP FORCE APPL
73	VEHNO	Num	3	VEHICLE NUMBER
74	VEHUSE	Num	3	VEHICLE SPECIAL USE

#	Variable	Type	Len	Label
75	VINJSER	Num	8	NUMBER SERIOUSLY INJURED IN THIS VEHICLE (AIS98 FORMAT)
76	VINJSER8	Num	8	NUMBER SERIOUSLY INJURED IN THIS VEHICLE (AIS08 FORMAT)
77	VINJURD8	Num	8	NUMBER INJURED IN THIS VEHICLE (AIS08 FORMAT)
78	VINJURED	Num	8	NUMBER INJURED IN THIS VEHICLE (AIS98 FORMAT)
79	VTREAT	Num	8	MAXIMUM TREATMENT IN THIS VEHICLE
80	WEATHER	Num	3	

6.6 Acc_desc (accident description) file

Contents (SAS Proc Contents, alphabetical order)

#	Variable	Type	Length	Label
1	CASENO	Num	8	CASE SEQUENCE NUMBER
2	CASEYEAR	Num	3	CASE YEAR
3	LINENO	Num	4	LINE NUMBER
4	PSU	Num	3	PRIMARY SAMPLING UNIT NUMBER
5	TEXT71	Char	80	SUMMARY TEXT

6.7 Event file

Contents (SAS Proc Contents, alphabetical order)

#	Variable	Type	Length	Label
1	ACCSEQ	Num	3	ACCIDENT EVENT SEQUENCE NUMBER
2	CASENO	Num	8	CASE SEQUENCE NUMBER
3	CASEYEAR	Num	3	CASE YEAR
4	CLASS1	Num	3	CLASS OF FIRST VEHICLE
5	CLASS2	Num	3	CLASS OF OTHER VEHICLE
6	GADEV1	Char	1	GENERAL AREA OF DAMAGE FIRST VEHICLE
7	GADEV2	Char	1	GENERAL AREA OF DAMAGE OTHER VEHICLE
8	OBJCONT	Num	3	OTHER VEHICLE NUMBER OR OBJECT CONTACTED
9	PSU	Num	3	PRIMARY SAMPLING UNIT NUMBER
10	VEHNO	Num	3	VEHICLE NUMBER

6.8 VE (vehicle exterior) file

Contents (SAS Proc Contents, alphabetical order)

#	Variable	Type	Length	Label
1	ACCSEQ1	Num	3	ACCIDENT EVENT SEQUENCE (HIGHEST)
2	ACCSEQ2	Num	3	ACCIDENT EVENT SEQUENCE (2ND HIGHEST)
3	ALTVEH	Num	3	MULTI-STAGE MANUFACTURED/CERT. ALT. VEH.
4	CASENO	Num	8	CASE SEQUENCE NUMBER
5	CASEYEAR	Num	3	CASE YEAR
6	DIRDAMW	Num	3	DIRECT DAMAGE WIDTH
7	DOCCDC	Num	3	CDCs DOCUMENTED BUT NOT CODED ON FILE?
8	DOF1	Num	3	DIRECTION OF FORCE (HIGHEST)
9	DOF2	Num	3	DIRECTION OF FORCE (2ND HIGHEST)
10	DVC1	Num	3	CRUSH PROFILE C1 (HIGHEST)
11	DVC2	Num	3	CRUSH PROFILE C2 (HIGHEST)
12	DVC3	Num	3	CRUSH PROFILE C3 (HIGHEST)
13	DVC4	Num	3	CRUSH PROFILE C4 (HIGHEST)
14	DVC5	Num	3	CRUSH PROFILE C5 (HIGHEST)
15	DVC6	Num	3	CRUSH PROFILE C6 (HIGHEST)
16	DVD	Num	3	CRUSH PROFILE D (HIGHEST)
17	DVL	Num	3	CRUSH PROFILE L (HIGHEST)
18	EXTENT1	Num	3	DEFORMATION EXTENT (HIGHEST)
19	EXTENT2	Num	3	DEFORMATION EXTENT (2ND HIGHEST)
20	FIRE	Num	3	FIRE OCCURRENCE
21	FIREORIG	Num	3	ORIGIN OF FIRE
22	FUELCAP1	Num	3	LOCATION OF FUEL TANK-1 FILLER CAP
23	FUELCAP2	Num	3	LOCATION OF FUEL TANK-2 FILLER CAP
24	FUELDAM1	Num	3	DAMAGE TO FUEL TANK-1
25	FUELDAM2	Num	3	DAMAGE TO FUEL TANK-2
26	FUELEAK1	Num	3	LEAKAGE LOCATION OF FUEL SYSTEM-1
27	FUELEAK2	Num	3	LEAKAGE LOCATION OF FUEL SYSTEM-2
28	FUELGT2	Num	3	EQUIPPED WITH MORE THAN TWO FUEL TANKS
29	FUELLOC1	Num	3	LOCATION OF FUEL TANK-1
30	FUELLOC2	Num	3	LOCATION OF FUEL TANK-2
31	FUELPRE1	Num	3	FUEL TANK-1 PRECRASH CONDITIONS
32	FUELPRE2	Num	3	FUEL TANK-2 PRECRASH CONDITIONS
33	FUELTNK1	Num	3	TYPE OF FUEL TANK-1
34	FUELTNK2	Num	3	TYPE OF FUEL TANK-2

#	Variable	Type	Length	Label
35	FUELTP1	Num	3	FUEL TYPE-1
36	FUELTP2	Num	3	FUEL TYPE-2
37	GAD1	Char	1	DEFORMATION LOCATION (HIGHEST)
38	GAD2	Char	1	DEFORMATION LOCATION (2ND HIGHEST)
39	OBJCONT1	Num	3	OBJECT CONTACTED (HIGHEST)
40	OBJCONT2	Num	3	OBJECT CONTACTED (2ND HIGHEST)
41	ORIGAVTW	Num	3	ORIGINAL AVERAGE TRACK WIDTH
42	PDOF1	Num	3	CLOCK DIRECTION FOR PDOF IN DEGREES (HIGHEST CDC)
43	PDOF2	Num	3	CLOCK DIRECTION FOR PDOF IN DEGREES (2ND HIGHEST CDC)
44	PSU	Num	3	PRIMARY SAMPLING UNIT NUMBER
45	SDVC1	Num	3	CRUSH PROFILE C1 (2ND HIGHEST)
46	SDVC2	Num	3	CRUSH PROFILE C2 (2ND HIGHEST)
47	SDVC3	Num	3	CRUSH PROFILE C3 (2ND HIGHEST)
48	SDVC4	Num	3	CRUSH PROFILE C4 (2ND HIGHEST)
49	SDVC5	Num	3	CRUSH PROFILE C5 (2ND HIGHEST)
50	SDVC6	Num	3	CRUSH PROFILE C6 (2ND HIGHEST)
51	SDVD	Num	3	CRUSH PROFILE D (2ND HIGHEST)
52	SDVL	Num	3	CRUSH PROFILE L (2ND HIGHEST)
53	SHL1	Char	1	SPECIFIC LONGITUDINAL LOCATION (HIGHEST)
54	SHL2	Char	1	SPECIFIC LONGITUDINAL LOC. (2ND HIGHEST)
55	SVL1	Char	1	SPECIFIC VERTICAL LOCATION (HIGHEST)
56	SVL2	Char	1	SPECIFIC VERTICAL LOCATION (2ND HIGHEST)
57	TDD1	Char	1	TYPE OF DAMAGE DISTRIBUTION (HIGHEST)
58	TDD2	Char	1	TYPE OF DAMAGE DISTRIBUTION(2ND HIGHEST)
59	TOWRES	Num	3	RESEARCHER ASSESSMNT VEHICLE DISPOSITION
60	UNDENDW	Num	3	UNDEFORMED END WIDTH
61	VEHNO	Num	3	VEHICLE NUMBER
62	WHEELBAS	Num	8	ORIGINAL WHEELBASE

6.9 VI (vehicle interior) file

Contents (SAS Proc Contents, alphabetical order)

#	Variable	Type	Length	Label
1	ADAPTEQ	Num	3	ADAPTIVE (ASSISTIVE) DRIVING EQUIPMENT
2	BOLSTDEF	Num	3	KNEE BOLSTER DEFORMED - OCCUPANT CONTACT
3	BOLSTYPE	Num	3	TYPE OF KNEE BOLSTER COVERING
4	CASENO	Num	8	CASE SEQUENCE NUMBER
5	CASEYEAR	Num	3	CASE YEAR
6	CDRIR1	Num	3	1ST DOMINANT CRUSH DIRECTION
7	CDRIR2	Num	3	2ND DOMINANT CRUSH DIRECTION
8	CDRIR3	Num	3	3RD DOMINANT CRUSH DIRECTION
9	CDRIR4	Num	3	4TH DOMINANT CRUSH DIRECTION
10	CDRIR5	Num	3	5TH DOMINANT CRUSH DIRECTION
11	CDRIR6	Num	3	6TH DOMINANT CRUSH DIRECTION
12	CDRIR7	Num	3	7TH DOMINANT CRUSH DIRECTION
13	CDRIR8	Num	3	8TH DOMINANT CRUSH DIRECTION
14	CDRIR9	Num	3	9TH DOMINANT CRUSH DIRECTION
15	CDRIR10	Num	3	10TH DOMINANT CRUSH DIRECTION
16	CDRIR11	Num	3	11TH DOMINANT CRUSH DIRECTION
17	CDRIR12	Num	3	12TH DOMINANT CRUSH DIRECTION
18	CDRIR13	Num	3	13TH DOMINANT CRUSH DIRECTION
19	CDRIR14	Num	3	14TH DOMINANT CRUSH DIRECTION
20	CDRIR15	Num	3	15TH DOMINANT CRUSH DIRECTION
21	CDRIR16	Num	3	16TH DOMINANT CRUSH DIRECTION
22	CDRIR17	Num	3	17TH DOMINANT CRUSH DIRECTION
23	CDRIR18	Num	3	18TH DOMINANT CRUSH DIRECTION
24	CDRIR19	Num	3	19TH DOMINANT CRUSH DIRECTION
25	CDRIR20	Num	3	20TH DOMINANT CRUSH DIRECTION
26	CDRIR21	Num	3	21ST DOMINANT CRUSH DIRECTION
27	CDRIR22	Num	3	22ND DOMINANT CRUSH DIRECTION
28	CDRIR23	Num	3	23RD DOMINANT CRUSH DIRECTION
29	CDRIR24	Num	3	24TH DOMINANT CRUSH DIRECTION
30	CDRIR25	Num	3	25TH DOMINANT CRUSH DIRECTION
31	COLMTELE	Num	3	TELESCOPING STEERING COLUMN ADJUSTMENT
32	COLMTILT	Num	3	TILT STEERING COLUMN ADJUSTMENT
33	COLUMTYP	Num	3	STEERING COLUMN TYPE

#	Variable	Type	Length	Label
34	FAILLF	Num	3	LF DAMAGE/FAILURE ASSOCIATED W
35	FAILLR	Num	3	LR DAMAGE/FAILURE - OPENING IN COLLISION
36	FAILRF	Num	3	RF DAMAGE/FAILURE - OPENING IN COLLISION
37	FAILRR	Num	3	RR DAMAGE/FAILURE - OPENING IN COLLISION
38	FAILTG	Num	3	TG DAMAGE/FAILURE - OPENING IN COLLISION
39	GLIMPBL	Num	3	BL GLAZING DAMAGE FROM IMPACT FORCES
40	GLIMPLF	Num	3	LF GLAZING DAMAGE FROM IMPACT FORCES
41	GLIMPLR	Num	3	LR GLAZING DAMAGE FROM IMPACT FORCES
42	GLIMPTH	Num	3	OTHER GLAZING DAMAGE FROM IMPACT FORCES
43	GLIMPRF	Num	3	RF GLAZING DAMAGE FROM IMPACT FORCES
44	GLIMPRR	Num	3	RR GLAZING DAMAGE FROM IMPACT FORCES
45	GLIMPRUF	Num	3	ROOF GLAZING DAMAGE FROM IMPACT FORCES
46	GLIMPWS	Num	3	WS GLAZING DAMAGE FROM IMPACT FORCES
47	GLOCCBL	Num	3	BL GLAZING DAMAGE FROM OCCUPANT CONTACT
48	GLOCCLF	Num	3	LF GLAZING DAMAGE FROM OCCUPANT CONTACT
49	GLOCCLR	Num	3	LR GLAZING DAMAGE FROM OCCUPANT CONTACT
50	GLOCCOTH	Num	3	OTHER GLAZING DAMAGE FROM OCC. CONTACT
51	GLOCCRF	Num	3	RF GLAZING DAMAGE FROM OCCUPANT CONTACT
52	GLOCCRR	Num	3	RR GLAZING DAMAGE FROM OCCUPANT CONTACT
53	GLOCCRUF	Num	3	ROOF GLAZING DAMAGE FROM OCC. CONTACT
54	GLOCCWS	Num	3	WS GLAZING DAMAGE FROM OCCUPANT CONTACT
55	GLOVOPEN	Num	3	DID GLOVE COMPARTMENT DOOR OPEN

#	Variable	Type	Length	Label
56	GLPREBL	Num	3	BL WINDOW PRECRASH GLAZING STATUS
57	GLPRELF	Num	3	LF WINDOW PRECRASH GLAZING STATUS
58	GLPRELR	Num	3	LR WINDOW PRECRASH GLAZING STATUS
59	GLPREOTH	Num	3	OTHER WINDOW PRECRASH GLAZING STATUS
60	GLPRERF	Num	3	RF WINDOW PRECRASH GLAZING STATUS
61	GLPRERR	Num	3	RR WINDOW PRECRASH GLAZING STATUS
62	GLPRERUF	Num	3	ROOF WINDOW PRECRASH GLAZING STATUS
63	GLPREWS	Num	3	WS WINDOW PRECRASH GLAZING STATUS
64	GLTYPBL	Num	3	BL TYPE OF WINDOW/WINDSHIELD GLAZING
65	GLTYPLF	Num	3	LF TYPE OF WINDOW/WINDSHIELD GLAZING
66	GLTYPLR	Num	3	LR TYPE OF WINDOW/WINDSHIELD GLAZING
67	GLTYPOTH	Num	3	OTHER TYPE OF WINDOW/WINDSHIELD GLAZING
68	GLTYPRF	Num	3	RF TYPE OF WINDOW/WINDSHIELD GLAZING
69	GLTYPRR	Num	3	RR TYPE OF WINDOW/WINDSHIELD GLAZING
70	GLTYPRUF	Num	3	ROOF TYPE OF WINDOW/WINDSHIELD GLAZING
71	GLTYPWS	Num	3	WS TYPE OF WINDOW/WINDSHIELD GLAZING
72	INCOMP1	Num	3	1ST INTRUDING COMPONENT
73	INCOMP2	Num	3	2ND INTRUDING COMPONENT
74	INCOMP3	Num	3	3RD INTRUDING COMPONENT
75	INCOMP4	Num	3	4TH INTRUDING COMPONENT
76	INCOMP5	Num	3	5TH INTRUDING COMPONENT
77	INCOMP6	Num	3	6TH INTRUDING COMPONENT
78	INCOMP7	Num	3	7TH INTRUDING COMPONENT
79	INCOMP8	Num	3	8TH INTRUDING COMPONENT
80	INCOMP9	Num	3	9TH INTRUDING COMPONENT
81	INCOMP10	Num	3	10TH INTRUDING COMPONENT
82	INCOMP11	Num	3	11TH INTRUDING COMPONENT
83	INCOMP12	Num	3	12TH INTRUDING COMPONENT
84	INCOMP13	Num	3	13TH INTRUDING COMPONENT
85	INCOMP14	Num	3	14TH INTRUDING COMPONENT

#	Variable	Type	Length	Label
86	INCOMP15	Num	3	15TH INTRUDING COMPONENT
87	INCOMP16	Num	3	16TH INTRUDING COMPONENT
88	INCOMP17	Num	3	17TH INTRUDING COMPONENT
89	INCOMP18	Num	3	18TH INTRUDING COMPONENT
90	INCOMP19	Num	3	19TH INTRUDING COMPONENT
91	INCOMP20	Num	3	20TH INTRUDING COMPONENT
92	INCOMP21	Num	3	21ST INTRUDING COMPONENT
93	INCOMP22	Num	3	22ND INTRUDING COMPONENT
94	INCOMP23	Num	3	23RD INTRUDING COMPONENT
95	INCOMP24	Num	3	24TH INTRUDING COMPONENT
96	INCOMP25	Num	3	25TH INTRUDING COMPONENT
97	INLOC1	Num	3	1ST LOCATION OF INTRUSION
98	INLOC2	Num	3	2ND LOCATION OF INTRUSION
99	INLOC3	Num	3	3RD LOCATION OF INTRUSION
100	INLOC4	Num	3	4TH LOCATION OF INTRUSION
101	INLOC5	Num	3	5TH LOCATION OF INTRUSION
102	INLOC6	Num	3	6TH LOCATION OF INTRUSION
103	INLOC7	Num	3	7TH LOCATION OF INTRUSION
104	INLOC8	Num	3	8TH LOCATION OF INTRUSION
105	INLOC9	Num	3	9TH LOCATION OF INTRUSION
106	INLOC10	Num	3	10TH LOCATION OF INTRUSION
107	INLOC11	Num	3	11TH LOCATION OF INTRUSION
108	INLOC12	Num	3	12TH LOCATION OF INTRUSION
109	INLOC13	Num	3	13TH LOCATION OF INTRUSION
110	INLOC14	Num	3	14TH LOCATION OF INTRUSION
111	INLOC15	Num	3	15TH LOCATION OF INTRUSION
112	INLOC16	Num	3	16TH LOCATION OF INTRUSION
113	INLOC17	Num	3	17TH LOCATION OF INTRUSION
114	INLOC18	Num	3	18TH LOCATION OF INTRUSION
115	INLOC19	Num	3	19TH LOCATION OF INTRUSION
116	INLOC20	Num	3	20TH LOCATION OF INTRUSION
117	INLOC21	Num	3	21ST LOCATION OF INTRUSION
118	INLOC22	Num	3	22ND LOCATION OF INTRUSION
119	INLOC23	Num	3	23RD LOCATION OF INTRUSION
120	INLOC24	Num	3	24TH LOCATION OF INTRUSION
121	INLOC25	Num	3	25TH LOCATION OF INTRUSION
122	INMAG1	Num	3	1ST MAGNITUDE OF INTRUSION
123	INMAG2	Num	3	2ND MAGNITUDE OF INTRUSION
124	INMAG3	Num	3	3RD MAGNITUDE OF INTRUSION

#	Variable	Type	Length	Label
125	INMAG4	Num	3	4TH MAGNITUDE OF INTRUSION
126	INMAG5	Num	3	5TH MAGNITUDE OF INTRUSION
127	INMAG6	Num	3	6TH MAGNITUDE OF INTRUSION
128	INMAG7	Num	3	7TH MAGNITUDE OF INTRUSION
129	INMAG8	Num	3	8TH MAGNITUDE OF INTRUSION
130	INMAG9	Num	3	9TH MAGNITUDE OF INTRUSION
131	INMAG10	Num	3	10TH MAGNITUDE OF INTRUSION
132	INMAG11	Num	3	11TH MAGNITUDE OF INTRUSION
133	INMAG12	Num	3	12TH MAGNITUDE OF INTRUSION
134	INMAG13	Num	3	13TH MAGNITUDE OF INTRUSION
135	INMAG14	Num	3	14TH MAGNITUDE OF INTRUSION
136	INMAG15	Num	3	15TH MAGNITUDE OF INTRUSION
137	INMAG16	Num	3	16TH MAGNITUDE OF INTRUSION
138	INMAG17	Num	3	17TH MAGNITUDE OF INTRUSION
139	INMAG18	Num	3	18TH MAGNITUDE OF INTRUSION
140	INMAG19	Num	3	19TH MAGNITUDE OF INTRUSION
141	INMAG20	Num	3	20TH MAGNITUDE OF INTRUSION
142	INMAG21	Num	3	21ST MAGNITUDE OF INTRUSION
143	INMAG22	Num	3	22ND MAGNITUDE OF INTRUSION
144	INMAG23	Num	3	23RD MAGNITUDE OF INTRUSION
145	INMAG24	Num	3	24TH MAGNITUDE OF INTRUSION
146	INMAG25	Num	3	25TH MAGNITUDE OF INTRUSION
147	ODOMETER	Num	3	ODOMETER READING
148	OPENLF	Num	3	LF DOOR, TAILGATE OR HATCH OPENING
149	OPENLR	Num	3	LR DOOR, TAILGATE OR HATCH OPENING
150	OPENRF	Num	3	RF DOOR, TAILGATE OR HATCH OPENING
151	OPENRR	Num	3	RR DOOR, TAILGATE OR HATCH OPENING
152	OPENTG	Num	3	TG DOOR, TAILGATE OR HATCH OPENING
153	PANELDAM	Num	3	INSTRUMENT PANEL DAMAGE - OCC.CONTACT
154	PASINTEG	Num	3	PASSENGER COMPARTMENT INTEGRITY
155	POSTINT	Num	3	POST CRASH INTEGRITY LOSS
156	PSU	Num	3	PRIMARY SAMPLING UNIT NUMBER
157	RDEFLOC	Num	3	LOCATION STEERING RIM/SPOKE DEFORMATION
158	RIMDEF	Num	3	STEERING RIM/SPOKE DEFORMATION
159	VEHNO	Num	3	VEHICLE NUMBER

6.10 OA (occupant assessment) file

Contents (SAS Proc Contents, alphabetical order)

#	Variable	Type	Length	Label
1	AGE	Num	3	AGE OF OCCUPANT
2	BELTANCH	Num	3	SHOULDER BELT UPPER ANCHORAGE ADJUSTMENT
3	BELTSOU	Num	3	PRIMARY SOURCE OF BELT USE DETERMINATION
4	CASENO	Num	8	CASE SEQUENCE NUMBER
5	CASEYEAR	Num	3	CASE YEAR
6	CHHARNES	Num	3	CHILD SAFETY SEAT HARNESS USAGE
7	CHMAKE	Num	3	CHILD SAFETY SEAT MAKE/MODEL
8	CHORIENT	Num	3	CHILD SAFETY SEAT ORIENTATION
9	CHOWUSED	Num	8	HOW CHILD SAFETY SEAT USED
10	CHSHIELD	Num	3	CHILD SAFETY SEAT SHIELD USAGE
11	CHTETHER	Num	3	CHILD SAFETY SEAT TETHER USAGE
12	CHTYPE	Num	3	TYPE OF CHILD SAFETY SEAT
13	CHUSED	Num	3	WAS CHILD SEAT USED?
14	EJCTAREA	Num	3	EJECTION AREA
15	EJCTMED	Num	3	EJECTION MEDIUM
16	EJECTION	Num	3	EJECTION
17	ENTRAP	Num	3	ENTRAPMENT
18	FETALDOA	Num	3	FETAL MORTALITY
19	HEADREST	Num	3	HEAD RESTRAINT TYPE/DAMAGE BY OCCUPANT
20	HEIGHT	Num	3	HEIGHT OF OCCUPANT
21	HOSPSTAY	Num	3	HOSPITAL STAY
22	INJSEV	Num	3	INJURY SEVERITY (POLICE RATING)
23	INTGREST	Num	3	INTEGRATED RESTRAINTS
24	MANAVAIL	Num	3	MANUAL BELT SYSTEM AVAILABILITY
25	MANFAIL	Num	3	MANUAL BELT FAILURE MODE DURING ACCIDENT
26	MANPROPR	Num	3	PROPER USE OF MANUAL BELTS
27	MANUSE	Num	3	MANUAL BELT SYSTEM USE
28	MEDFACIL	Num	3	TYPE MEDICAL FACILITY INITIAL TREATMENT
29	MEDSTA	Num	3	MEDIUM STATUS (PRIOR TO IMPACT)
30	OCCMOBIL	Num	3	OCCUPANT MOBILITY
31	OCCNO	Num	3	OCCUPANT NUMBER
32	OCCRACE	Num	3	OCCUPANTS RACE

#	Variable	Type	Length	Label
33	OCETHNIC	Num	3	OCCUPANTS ETHNICITY
34	POSTURE	Num	3	OCCUPANT'S POSTURE
35	PSU	Num	3	PRIMARY SAMPLING UNIT NUMBER
36	ROLE	Num	3	OCCUPANT'S ROLE
37	ROLLPROT	Num	3	ROLLOVER PROTECTION
38	SEATPERF	Num	3	SEAT PERFORMANCE (THIS POSITION)
39	SEATPOS	Num	3	OCCUPANT'S SEAT POSITION
40	SEATRACK	Num	3	SEAT TRACK ADJUSTED POSITION PRIOR TO IMPACT
41	SEATTYPE	Num	3	SEAT TYPE (THIS OCCUPANT POSITION)
42	SEX	Num	3	OCCUPANT'S SEX
43	STBACINC	Num	3	SEAT BACK INCLINE PRIOR AND POST IMPACT
44	STORIENT	Num	3	SEAT ORIENTATION (THIS OCCUPANT POS.)
45	TREATMNT	Num	3	TREATMENT - MORTALITY
46	VEHNO	Num	3	VEHICLE NUMBER
47	WEIGHT	Num	3	OCCUPANT'S WEIGHT
48	WORKDAYS	Num	3	WORKING DAYS LOST

Please reference the appropriate NASS-CDS Analytical User's Manual for details on these variables.

Appendix A

References

1. National Academy of Sciences (1985) Injury in America – A Continuing Public Health Problem. Washington, D.C. National Academy Press (pg. 62)
2. Schneider, Lawrence W.; Rupp, Jonathan D.; Scarboro, Mark; Pintar, Frank; Arbogast, Kristy B.; Rudd, Rodney W.; Sochor, Mark R.; Stitzel, Joel; Sherwood, Chris; MacWilliams, Joel B.; Halloway, Dale; Ridella, Stephen and Eppinger, Rolf (2011) 'BioTab-A New Method for Analyzing and Documenting Injury Causation in Motor-Vehicle Crashes', Traffic Injury Prevention, 12: 3, 256 – 265
3. Association for the Advancement of Automotive Medicine. The Abbreviated Injury Scale – AIS 90/Update 1998. Des Plaines, Ill: 1998
4. Association for the Advancement of Automotive Medicine. The Abbreviated Injury Scale – AIS 2005/Update 2008. Des Plaines, Ill: 2008
5. American Medical Association. International Classification of Diseases 9th Revision Clinical Modification. 2003

Appendix B

NASS-CDS variable availability in the CIREN version of the Acc_desc, Event, GV, VE, VI, and OA files. Since variables in NASS-CDS have changed over time, the year-specific columns indicate whether a variable was present in the NASS-CDS data set for each data year included in the CIREN public data set.

[illegible]

[illegible]

[illegible]

	CIREN	NASS-CDS	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014-2016
TOWPAR	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
TRAFCONT	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
TRAFFLOW	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
TRANSTAT	Y	Y	N	N	Y	Y	Y	Y	Y	Y	Y	Y
TRAVELSP	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
TRCTLFCT	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
TRIPLOC	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
VAIS	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
VAIS08	N	Y	N	N	N	N	N	Y	Y	Y	Y	Y
VEHNO	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
VEHTYPE	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
VEHUSE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
VEHWGT	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
VERSION	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
VIN	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
VINAMOD	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
VINBT	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
VINJSER	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
VINJSER8	Y	Y	N	N	N	N	N	Y	Y	Y	Y	Y
VINJURD8	Y	Y	N	N	N	N	N	Y	Y	Y	Y	Y
VINJURED	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
VINLNGTH	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
VINMAKE	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
VINMODYR	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
VINO	N	Y	Y	Y	Y	Y	N	N	N	N	N	N
VTREAT	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
WEATHER	Y	Y	Y	Y	N	N	N	N	N	N	N	N
WGTCDTR	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
WHLDRWHL	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
ABELTAVL	N	Y	Y	Y	Y	Y	Y	N	N	N	N	N
ABELTUSE	N	Y	Y	Y	Y	Y	Y	N	N	N	N	N
ABELTYPE	N	Y	Y	Y	Y	Y	Y	N	N	N	N	N
ABLTFAIL	N	Y	Y	Y	Y	Y	Y	N	N	N	N	N
ABLTPROP	N	Y	Y	N	N	N	N	N	N	N	N	N
AGE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
BAGAVAIL	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
BAGAVOTH	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
BAGAVRPT	N	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

OA

[illegible]

[illegible]

[illegible]

[illegible]

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